

Visión por Computador

Versión 2026

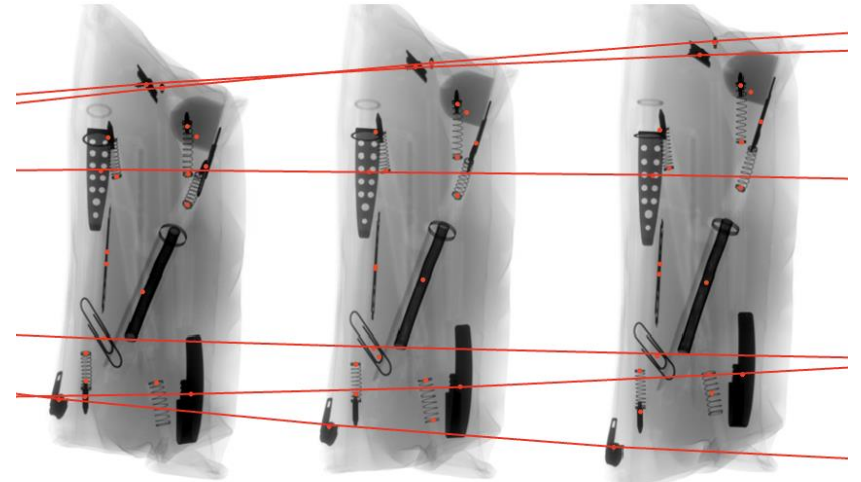


Múltiples Vistas

(CAPÍTULO 2 – CV02F)

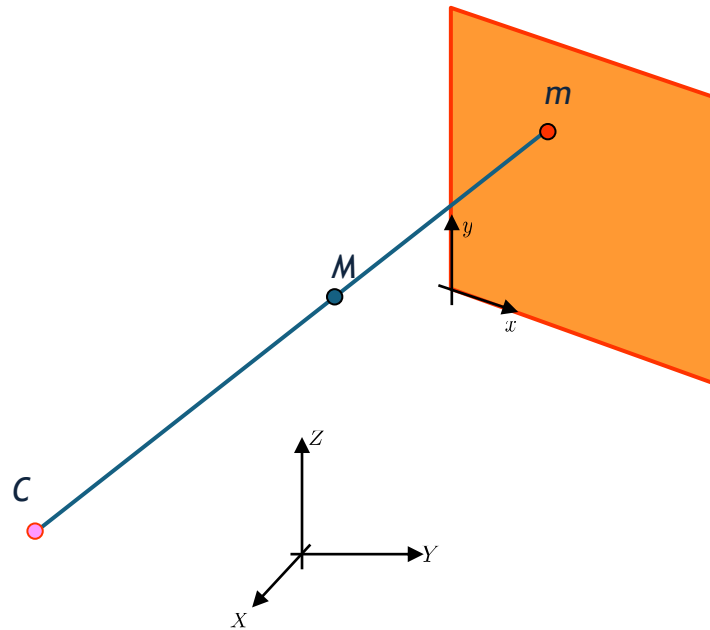
Domingo Mery 

Ciencia de la Computación
Universidad Católica – Chile
<http://domingomery.ing.uc.cl>



Reconstrucción 3D

3D > 2D Projection:

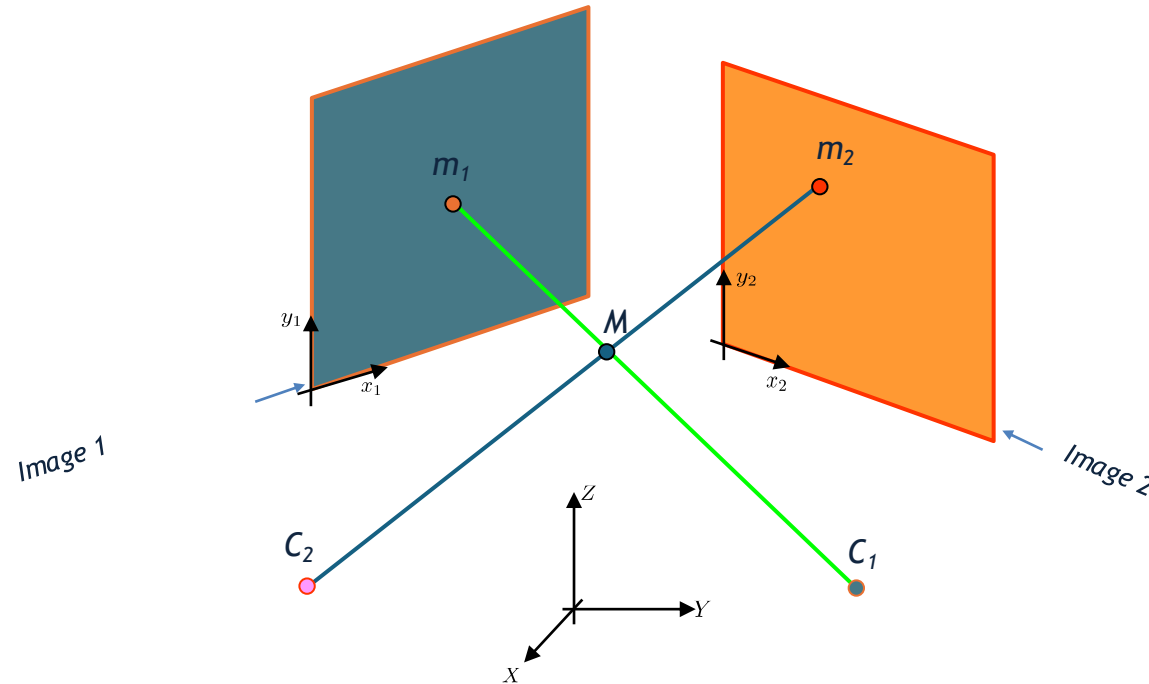


$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\lambda \mathbf{m} = \mathbf{P} \mathbf{M}$$

[3x1] [3x4] [4x1]

3D Reconstruction for two views:



Problem:

Given m_1 and m_2 (and projection matrices A and B)
find M

$$\lambda_1 \mathbf{m}_1 = \mathbf{A}\mathbf{M}$$

$$\lambda_2 \mathbf{m}_2 = \mathbf{B}\mathbf{M}$$

$$\lambda_1 \mathbf{m}_1 = \mathbf{A} \mathbf{M}$$

$$\lambda_2 \mathbf{m}_2 = \mathbf{B} \mathbf{M}$$

Known: a_{ij} , b_{ij} , x_1 , y_1 , x_2 , y_2 .

Unknown: λ_1 , λ_2 , X , Y , Z .

$$\lambda_1 \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\lambda_2 \begin{bmatrix} x_2 \\ y_2 \\ 1 \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\lambda_1 \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\lambda_1 = a_{31}X + a_{32}Y + a_{33}Z + a_{34}$$

$$\lambda_1 \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\lambda_1 = a_{31}X + a_{32}Y + a_{33}Z + a_{34}$$

$$(a_{31}X + a_{32}Y + a_{33}Z + a_{34})x_1 = a_{11}X + a_{12}Y + a_{13}Z + a_{14}$$

$$(a_{31}X + a_{32}Y + a_{33}Z + a_{34})y_1 = a_{21}X + a_{22}Y + a_{23}Z + a_{24}$$

$$a_{31}Xx_1 + a_{32}Yx_1 + a_{33}Zx_1 + a_{34}x_1 = a_{11}X + a_{12}Y + a_{13}Z + a_{14}$$

$$a_{31}Xy_1 + a_{32}Yy_1 + a_{33}Zy_1 + a_{34}y_1 = a_{21}X + a_{22}Y + a_{23}Z + a_{24}$$

$$a_{31}Xx_1 - a_{11}X + a_{32}Yx_1 - a_{12}Y + a_{33}Zx_1 - a_{13}Z = a_{14} - a_{34}x_1$$

$$a_{31}Xy_1 - a_{21}X + a_{32}Yy_1 - a_{22}Y + a_{33}Zy_1 - a_{23}Z = a_{24} - a_{34}y_1$$

$$(a_{31}x_1 - a_{11})X + (a_{32}x_1 - a_{12})Y + (a_{33}x_1 - a_{13})Z = a_{14} - a_{34}x_1$$

$$(a_{31}y_1 - a_{21})X + (a_{32}y_1 - a_{22})Y + (a_{33}y_1 - a_{23})Z = a_{24} - a_{34}y_1$$

$$(a_{31}X + a_{32}Y + a_{33}Z + a_{34})x_1 = a_{11}X + a_{12}Y + a_{13}Z + a_{14}$$

$$(a_{31}X + a_{32}Y + a_{33}Z + a_{34})y_1 = a_{21}X + a_{22}Y + a_{23}Z + a_{24}$$

$$\lambda_1 \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$(a_{31}x_1 - a_{11})X + (a_{32}x_1 - a_{12})Y + (a_{33}x_1 - a_{13})Z = a_{14} - a_{34}x_1$$

$$(a_{31}y_1 - a_{21})X + (a_{32}y_1 - a_{22})Y + (a_{33}y_1 - a_{23})Z = a_{14} - a_{24}y_1$$

$$\begin{bmatrix} q_{11} & q_{12} & q_{13} \\ q_{21} & q_{22} & q_{23} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$\lambda_2 \begin{bmatrix} x_2 \\ y_2 \\ 1 \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\begin{aligned} (b_{31}x_2 - b_{11})X + (b_{32}x_2 - b_{12})Y + (b_{33}x_2 - b_{13})Z &= b_{14} - b_{34}x_2 \\ (b_{31}y_2 - b_{21})X + (b_{32}y_2 - b_{22})Y + (b_{33}y_2 - b_{23})Z &= b_{14} - b_{24}y_2 \end{aligned}$$

$$\begin{bmatrix} q_{31} & q_{32} & q_{33} \\ q_{41} & q_{42} & q_{43} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} r_3 \\ r_4 \end{bmatrix}$$

$$\lambda_1 \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \longrightarrow \begin{bmatrix} q_{11} & q_{12} & q_{13} \\ q_{21} & q_{22} & q_{23} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$\lambda_2 \begin{bmatrix} x_2 \\ y_2 \\ 1 \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \longrightarrow \begin{bmatrix} q_{31} & q_{32} & q_{33} \\ q_{41} & q_{42} & q_{43} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} r_3 \\ r_4 \end{bmatrix}$$

$$\begin{bmatrix} q_{11} & q_{12} & q_{13} \\ q_{21} & q_{22} & q_{23} \\ q_{31} & q_{32} & q_{33} \\ q_{41} & q_{42} & q_{43} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{bmatrix}$$

$$\text{QM} = \mathbf{r}$$

Known
Unknown

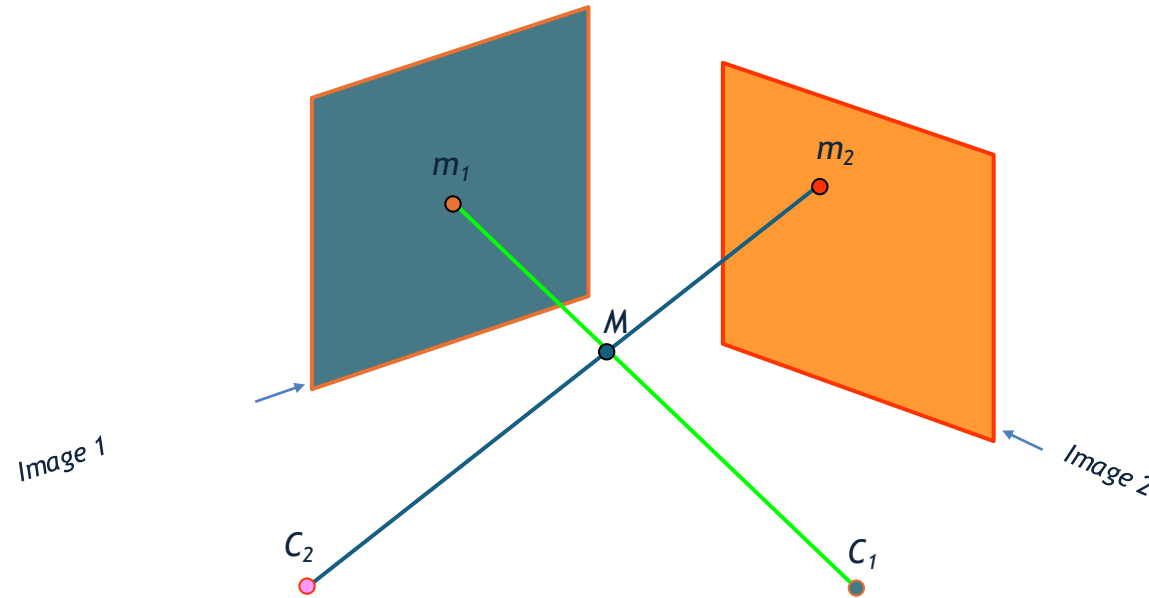
$$\overset{4 \times 3}{\mathbf{Q}} \overset{3 \times 1}{\mathbf{M}} = \overset{4 \times 1}{\mathbf{r}}$$

$$\mathbf{Q}^T \mathbf{Q} \mathbf{M} = \mathbf{Q}^T \mathbf{r}$$

$$[\mathbf{Q}^T \mathbf{Q}]^{-1} \mathbf{Q}^T \mathbf{Q} \mathbf{M} = [\mathbf{Q}^T \mathbf{Q}]^{-1} \mathbf{Q}^T \mathbf{r}$$

$$\mathbf{M} = [\mathbf{Q}^T \mathbf{Q}]^{-1} \mathbf{Q}^T \mathbf{r}$$

3D Reconstruction for two views:



Problem:

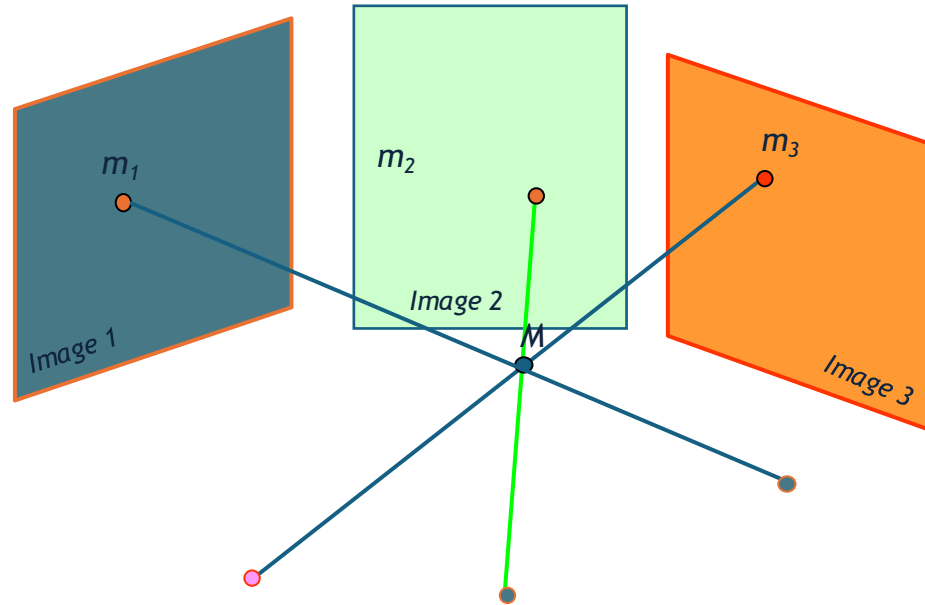
Given m_1 and m_2 (and projection matrices A and B)
find M

$$\mathbf{M} = [\mathbf{Q}^T \mathbf{Q}]^{-1} \mathbf{Q}^T \mathbf{r}$$

$$\lambda_1 \mathbf{m}_1 = \mathbf{A} \mathbf{M}$$

$$\lambda_2 \mathbf{m}_2 = \mathbf{B} \mathbf{M}$$

3D Reconstruction for 3 views:



$$\lambda_1 \mathbf{m}_1 = \mathbf{A}\mathbf{M}$$

$$\lambda_2 \mathbf{m}_2 = \mathbf{B}\mathbf{M}$$

$$\lambda_3 \mathbf{m}_3 = \mathbf{C}\mathbf{M}$$

Problem:

Given m_1 , m_2 and m_3 (and projection matrices A , B and C)
find M

$$\begin{aligned}
\lambda_1 \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} &= \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \longrightarrow \begin{bmatrix} q_{11} & q_{12} & q_{13} \\ q_{21} & q_{22} & q_{23} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} \\
\lambda_2 \begin{bmatrix} x_2 \\ y_2 \\ 1 \end{bmatrix} &= \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \longrightarrow \begin{bmatrix} q_{31} & q_{32} & q_{33} \\ q_{41} & q_{42} & q_{43} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} r_3 \\ r_4 \end{bmatrix} \\
\lambda_3 \begin{bmatrix} x_3 \\ y_3 \\ 1 \end{bmatrix} &= \begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} \\ c_{21} & c_{22} & c_{23} & c_{24} \\ c_{31} & c_{32} & c_{33} & c_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \longrightarrow \begin{bmatrix} q_{51} & q_{52} & q_{53} \\ q_{61} & q_{62} & q_{63} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} r_5 \\ r_6 \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
&\mathbf{QM} = \mathbf{r} \\
&\mathbf{M} = [\mathbf{Q}^\top \mathbf{Q}]^{-1} \mathbf{Q}^\top \mathbf{r}
\end{aligned}$$

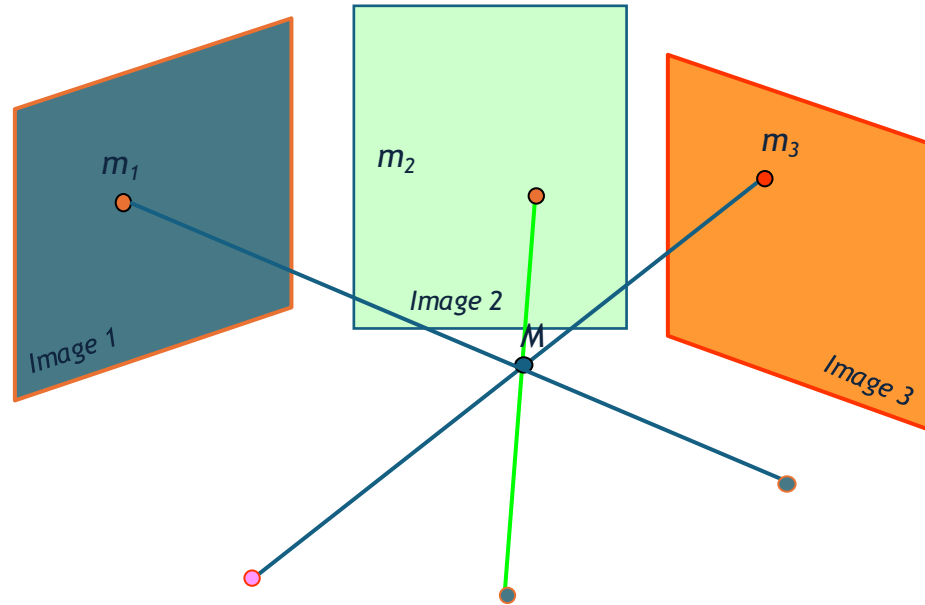
$\begin{bmatrix} q_{11} & q_{12} & q_{13} \\ q_{21} & q_{22} & q_{23} \\ q_{31} & q_{32} & q_{33} \\ q_{41} & q_{42} & q_{43} \\ q_{51} & q_{52} & q_{53} \\ q_{61} & q_{62} & q_{63} \end{bmatrix}$

$\downarrow$

$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \\ r_6 \end{bmatrix}$

$\longleftarrow$

3D Reconstruction for 3 views:



$$\lambda_1 \mathbf{m}_1 = \mathbf{A}\mathbf{M}$$

$$\lambda_2 \mathbf{m}_2 = \mathbf{B}\mathbf{M}$$

$$\lambda_3 \mathbf{m}_3 = \mathbf{C}\mathbf{M}$$

Problem:

Given m_1 , m_2 and m_3 (and projection matrices A , B and C)
find M

$$\mathbf{M} = [\mathbf{Q}^T \mathbf{Q}]^{-1} \mathbf{Q}^T \mathbf{r}$$

3D Reconstruction for n views:

$$\mathbf{M} = [\mathbf{Q}^T \mathbf{Q}]^{-1} \mathbf{Q}^T \mathbf{r}$$

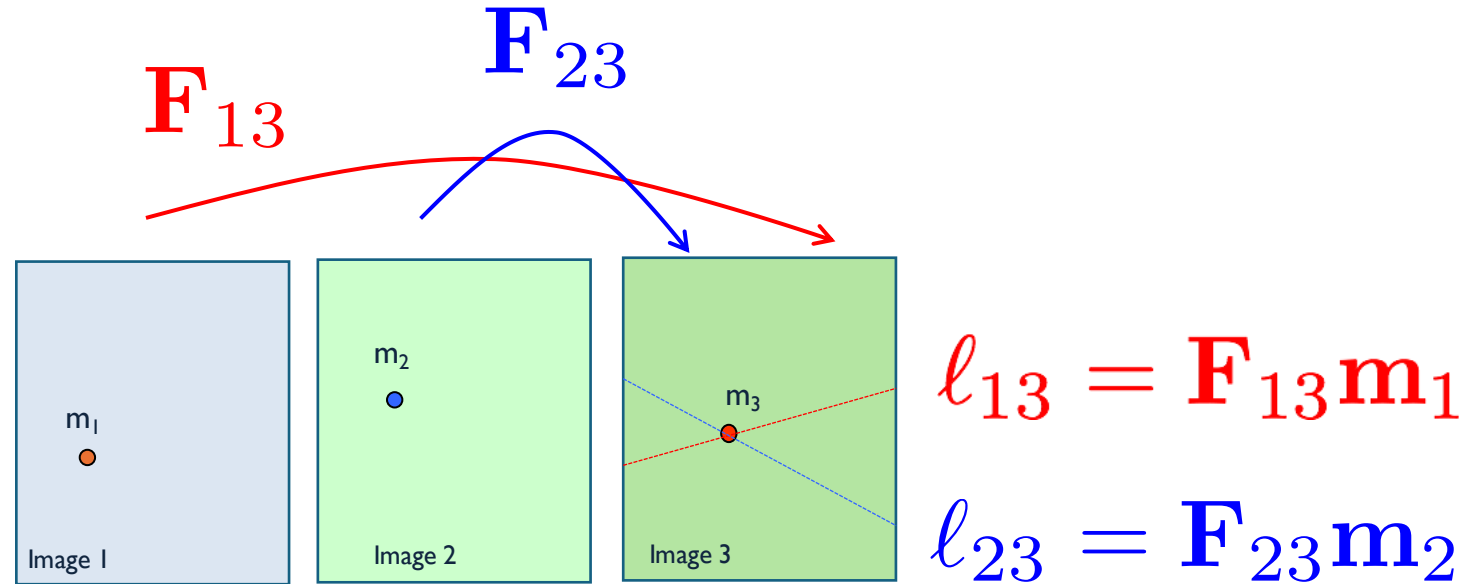
Size of Q: $2n \times 3$

Size of r: $2n \times 1$

Size of M: 3×1

Geometría Trifocal

Epipolar Geometry for 3 views:



$$\mathbf{F}_{13} = \text{fundamental}(\mathbf{A}, \mathbf{C})$$

$$\mathbf{F}_{23} = \text{fundamental}(\mathbf{B}, \mathbf{C})$$

Problem:

Given $\mathbf{m}_1, \mathbf{m}_2$ (and projection matrices $\mathbf{A}, \mathbf{B}$ and $\mathbf{C}$)
find $\mathbf{m}_3$

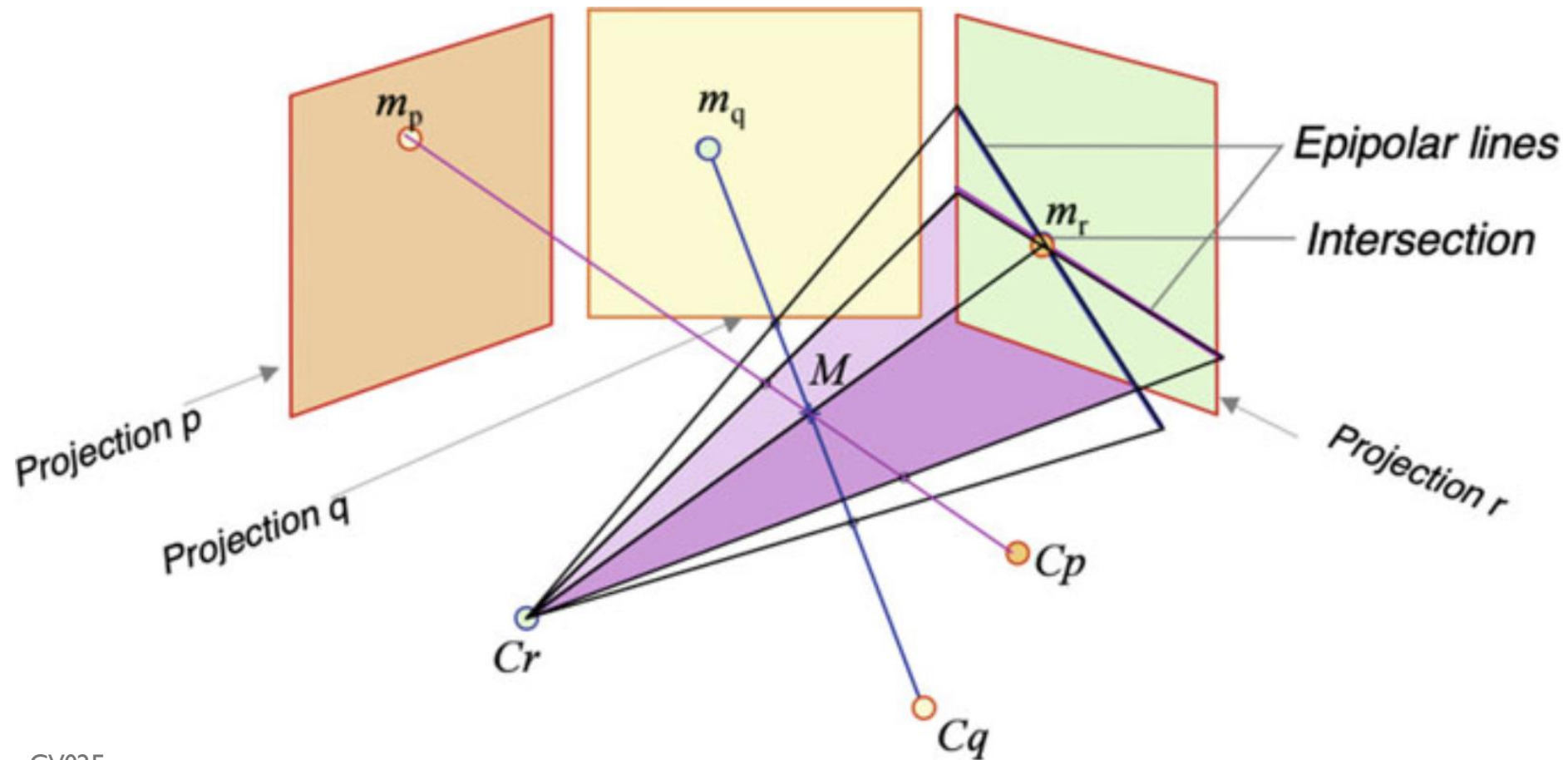
$$\mathbf{m}_3 = \ell_{13} \times \ell_{23}$$

$$\lambda_1 \mathbf{m}_1 = \mathbf{A}\mathbf{M}$$

$$\lambda_2 \mathbf{m}_2 = \mathbf{B}\mathbf{M}$$

$$\lambda_3 \mathbf{m}_3 = \mathbf{C}\mathbf{M}$$

$$\begin{cases} \lambda_p \mathbf{m}_p = \mathbf{P}_p \mathbf{M} := \mathbf{A} \mathbf{M} \\ \lambda_q \mathbf{m}_q = \mathbf{P}_q \mathbf{M} := \mathbf{B} \mathbf{M} \\ \lambda_r \mathbf{m}_r = \mathbf{P}_r \mathbf{M} := \mathbf{C} \mathbf{M} \end{cases}$$



$$\begin{cases} \lambda_p \mathbf{m}_p = \mathbf{P}_p \mathbf{M} := \mathbf{A} \mathbf{M} \\ \lambda_q \mathbf{m}_q = \mathbf{P}_q \mathbf{M} := \mathbf{B} \mathbf{M} \\ \lambda_r \mathbf{m}_r = \mathbf{P}_r \mathbf{M} := \mathbf{C} \mathbf{M} \end{cases} \quad \text{---} \downarrow$$

$$\underbrace{\begin{bmatrix} \mathbf{a}_1 & x_p & 0 & 0 \\ \mathbf{a}_2 & y_p & 0 & 0 \\ \mathbf{a}_3 & 1 & 0 & 0 \\ \mathbf{b}_1 & 0 & x_q & 0 \\ \mathbf{b}_2 & 0 & y_q & 0 \\ \mathbf{b}_3 & 0 & 1 & 0 \\ \mathbf{c}_1 & 0 & 0 & x_r \\ \mathbf{c}_2 & 0 & 0 & y_r \\ \mathbf{c}_3 & 0 & 0 & 1 \end{bmatrix}}_{\mathbf{G}} \underbrace{\begin{bmatrix} \mathbf{M} \\ -\lambda_p \\ -\lambda_q \\ -\lambda_r \end{bmatrix}}_{\mathbf{v}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad (3.74)$$

where $\mathbf{a}_i$, $\mathbf{b}_i$ and $\mathbf{c}_i$ denotes the i th row of matrix $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ respectively. If m_p , m_q and m_r are corresponding points, then there must be a nontrivial solution for $\mathbf{v}$. It follows that the rank of the 9×7 matrix $\mathbf{G}$ must be at most 6. In other words, all 7×7 submatrices have vanishing determinants.

$$\begin{cases} D_1 = (x_r \mathbf{T}^{13} - x_r x_q \mathbf{T}^{33} + x_q \mathbf{T}^{31} - \mathbf{T}^{11}) \mathbf{m}_p = 0 \\ D_2 = (y_r \mathbf{T}^{13} - y_r x_q \mathbf{T}^{33} + x_q \mathbf{T}^{32} - \mathbf{T}^{12}) \mathbf{m}_p = 0 \\ D_3 = (x_r \mathbf{T}^{23} - x_r y_q \mathbf{T}^{33} + y_q \mathbf{T}^{31} - \mathbf{T}^{21}) \mathbf{m}_p = 0 \\ D_4 = (y_r \mathbf{T}^{23} - y_r y_q \mathbf{T}^{33} + y_q \mathbf{T}^{32} - \mathbf{T}^{22}) \mathbf{m}_p = 0 \end{cases}, \quad (3.75)$$

where

$$\mathbf{T}^{jk} = [T_1^{jk} \ T_2^{jk} \ T_3^{jk}],$$

and

$$T_i^{jk} = (-1)^{i+1} \left| \begin{array}{c} \sim \mathbf{a}_i \\ \mathbf{b}_j \\ \mathbf{c}_k \end{array} \right| = \quad \text{for } i, j, k = 1, 2, 3, \quad (3.76)$$

where $\sim \mathbf{a}_i$ means the matrix $\mathbf{A}$ without row i . The elements T_i^{jk} are called the *trifocal tensors* for the images p , q and r [10, 11]. For the canonical form $\mathbf{A} = [\mathbf{I} \mid \mathbf{0}]$, the trifocal tensors may be easily obtained by

$$T_i^{jk} = b_{ji} c_{k4} - b_{j4} c_{ki} \quad \text{for } i, j, k = 1, 2, 3. \quad (3.77)$$

The *reprojection* of m_r , i.e., the coordinates $\hat{x}_r$ and $\hat{y}_r$ obtained from the points m_p and m_q , may be simply estimated from the trilinearities (3.75):

$$\lambda \hat{\mathbf{m}}_r = (\mathbf{T}^1 - x_q \mathbf{T}^3) \mathbf{m}_p = (\mathbf{T}^2 - y_q \mathbf{T}^3) \mathbf{m}_p, \quad (3.78)$$

where λ is a scale factor, $\hat{\mathbf{m}}_r = [\hat{x}_r \ \hat{y}_r \ 1]^\top$, and $\mathbf{T}^j$ is a 3×3 matrix with the (k, i) -element equal to T_i^{jk} .

In practice, given two corresponding points $\mathbf{m}_p$ and $\mathbf{m}_q$, the third one $\mathbf{m}_r$ can be considered as the corresponding point in third view, if the Euclidean distance between $\mathbf{m}_r$ and its reprojection $\hat{\mathbf{m}}_r$ is smaller than a small number ε :

$$d_3 = \|\hat{\mathbf{m}}_r - \mathbf{m}_r\| < \varepsilon. \quad (3.79)$$

Example

INPUT I

Figure q



Figure r



Application: Bagagge Screening

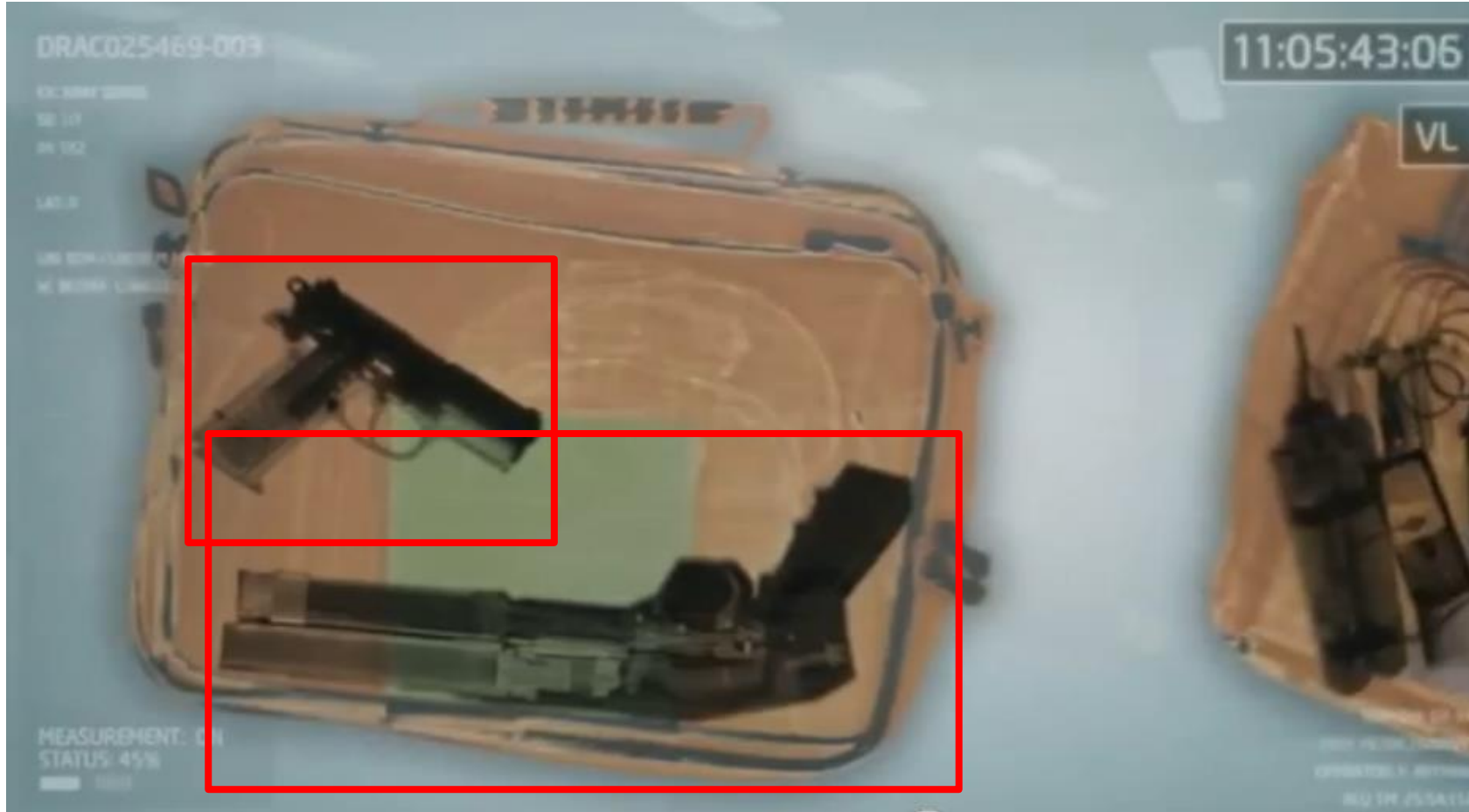
X-ray testing: **Baggage screening**

X-ray testing: **Baggage screening**



X-ray testing: Baggage screening

X-ray testing: **Baggage screening**



Our goal:

- 1) Detection of objects of interest
- 2) Views of each object in all frames



X-ray testing: **Baggage screening**

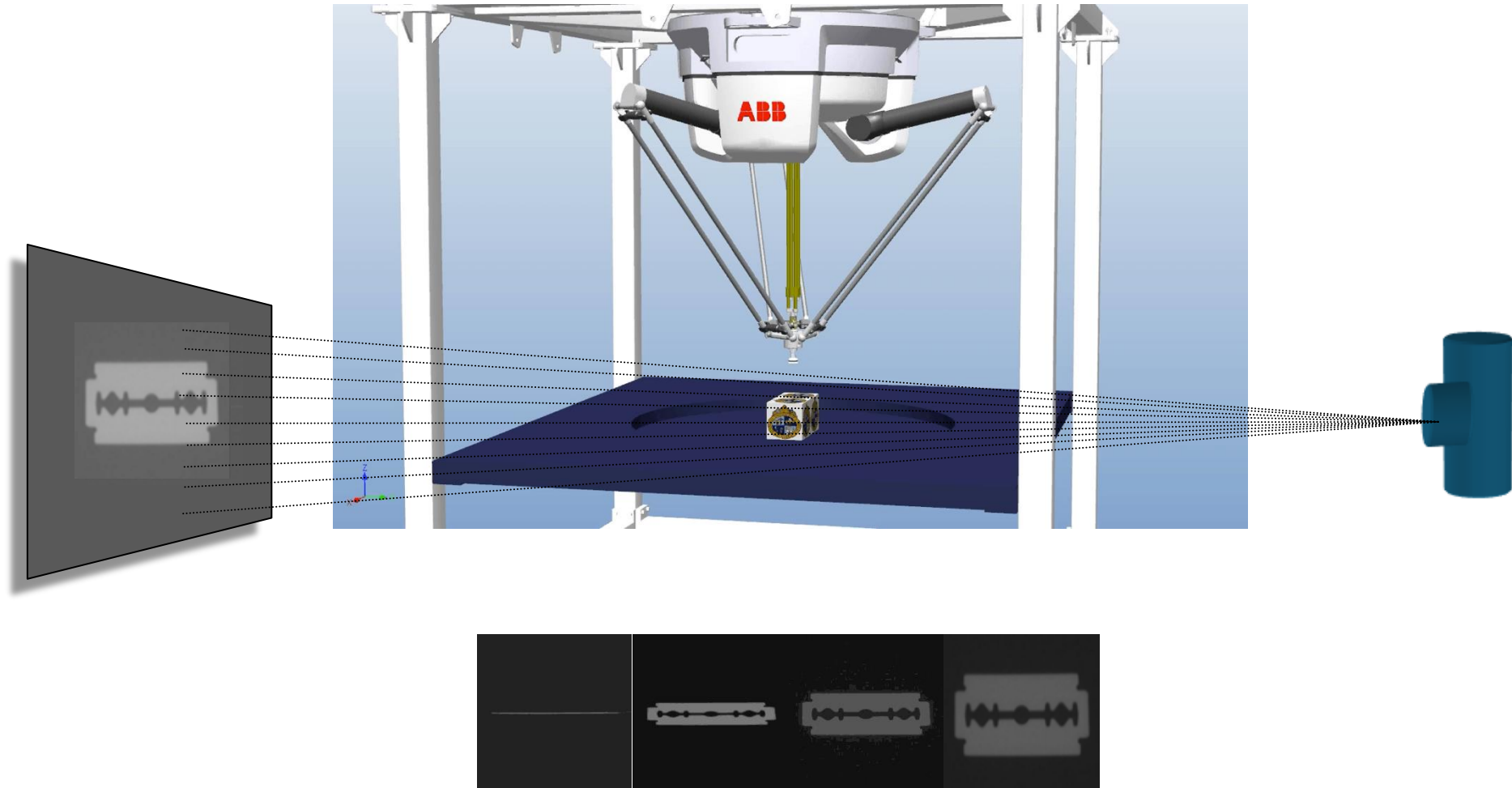
Key ideas of our methods:

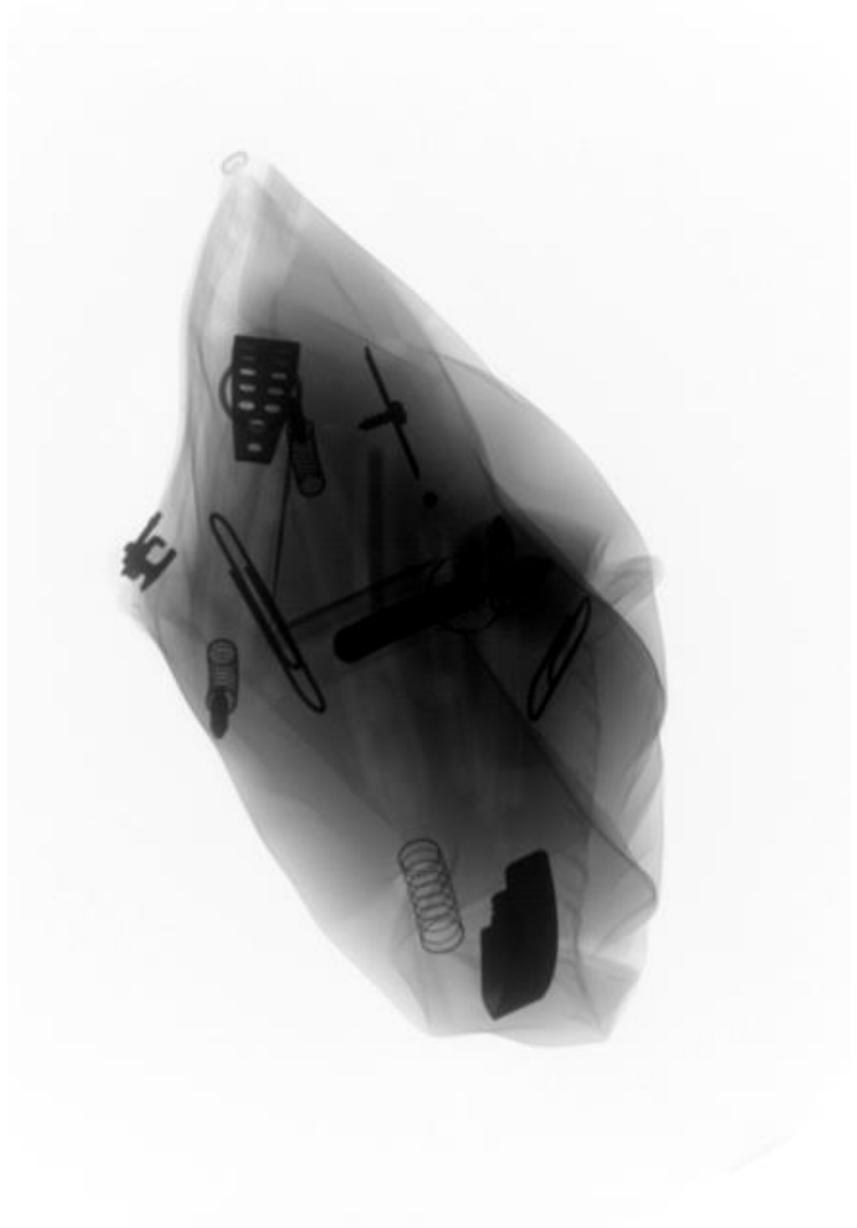
- **Computer vision**
- **Multiple views analysis**
- **Active vision**



X-ray Testing: Multiple view analysis

X-ray Testing: Multiple view analysis





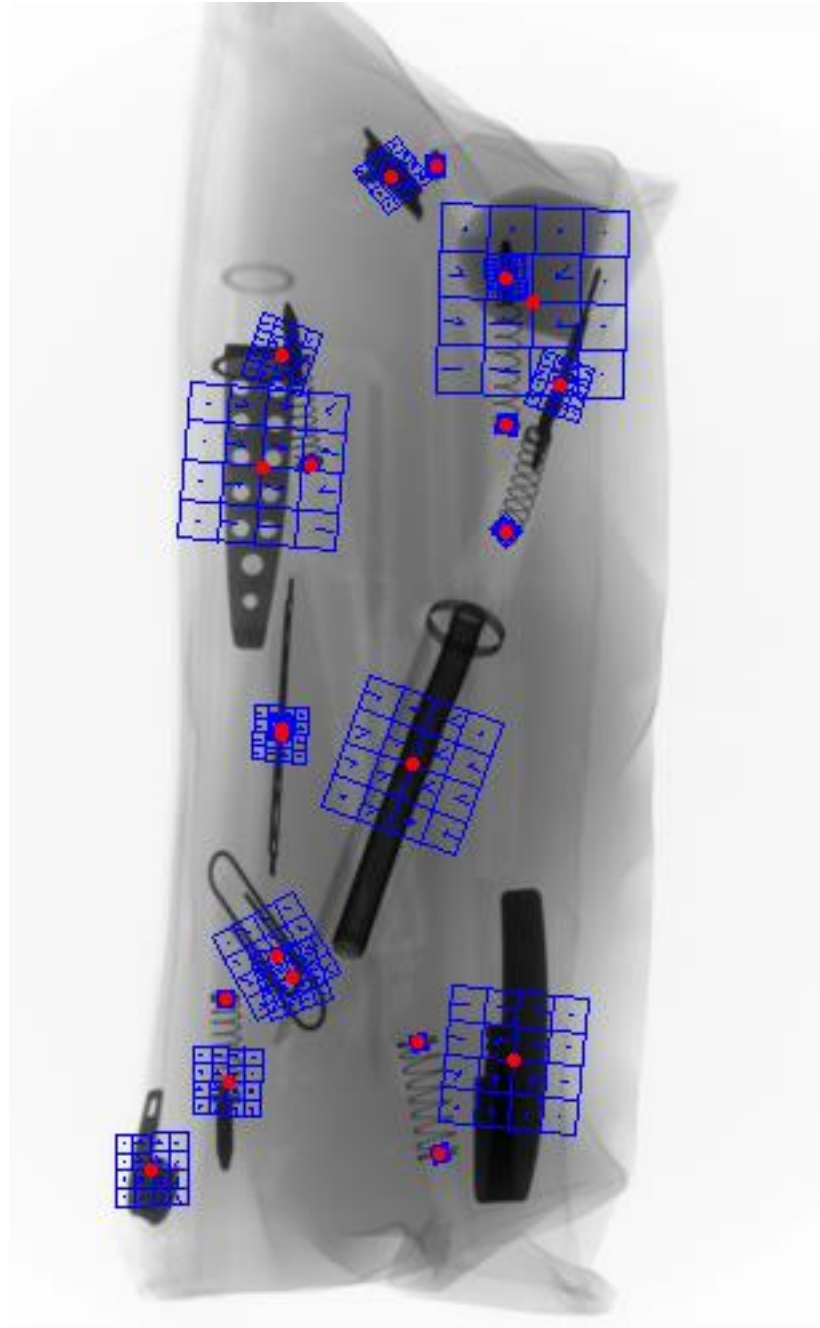




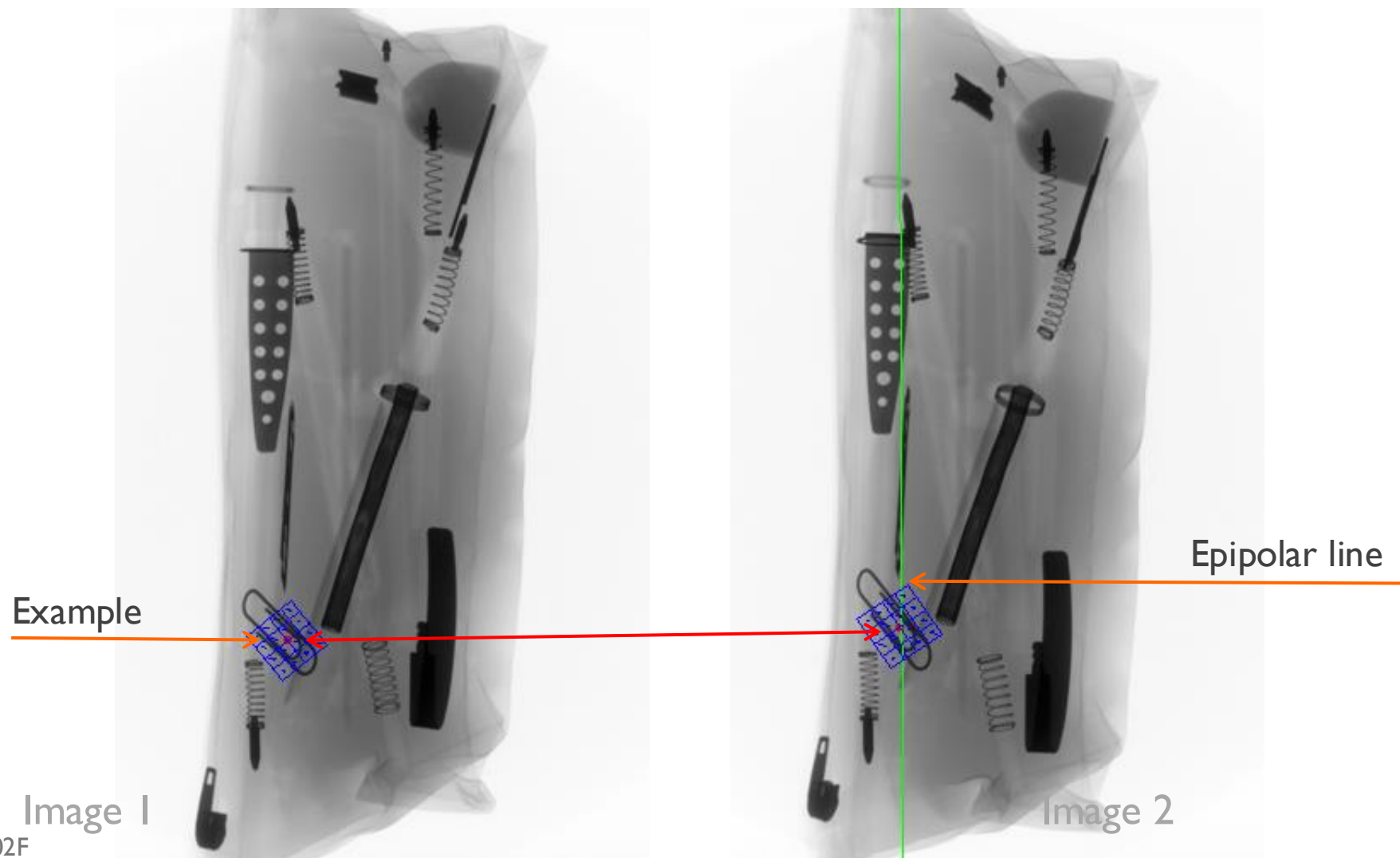




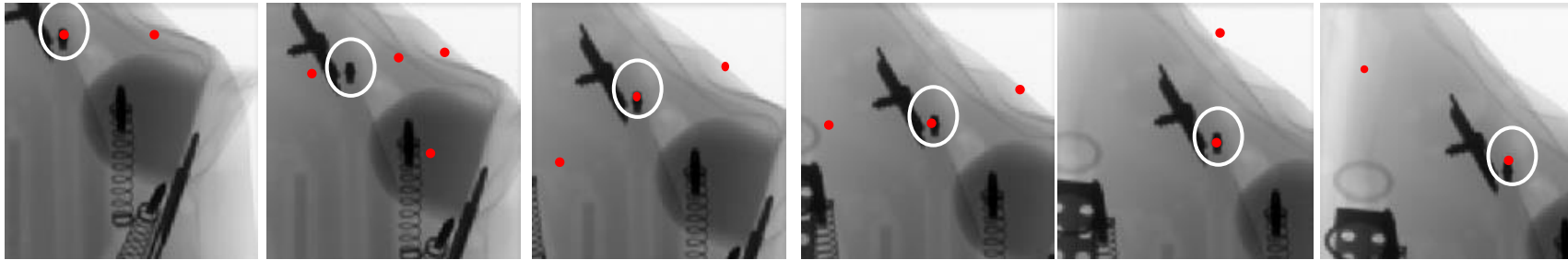
SIFT description for each
segmented region



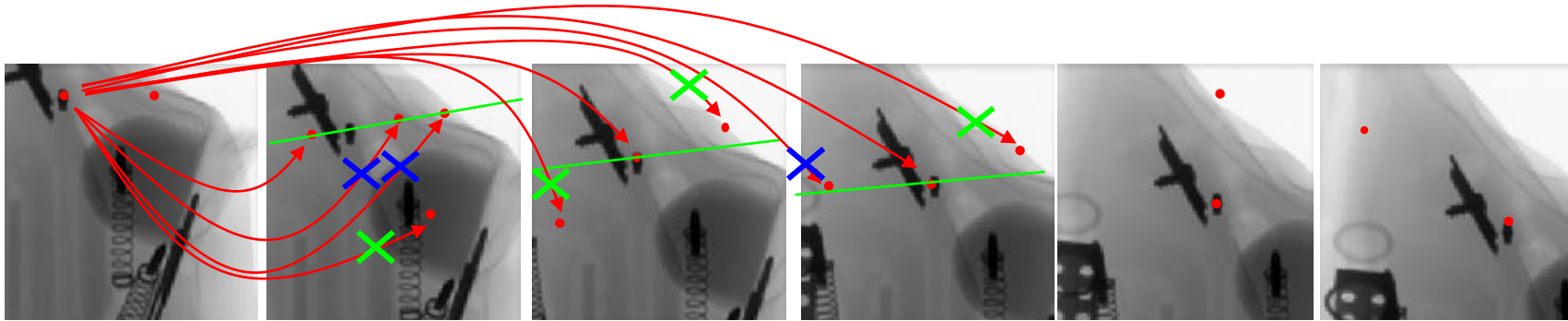
Tracking in 2 views



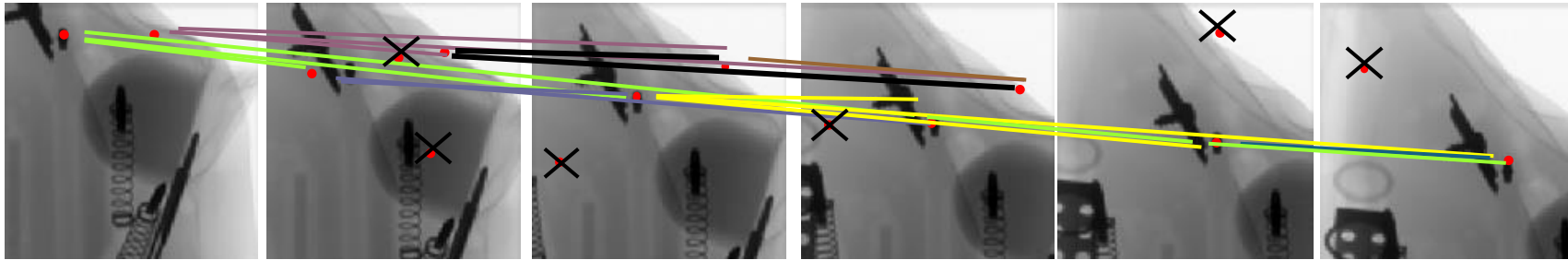
Tracking algorithm



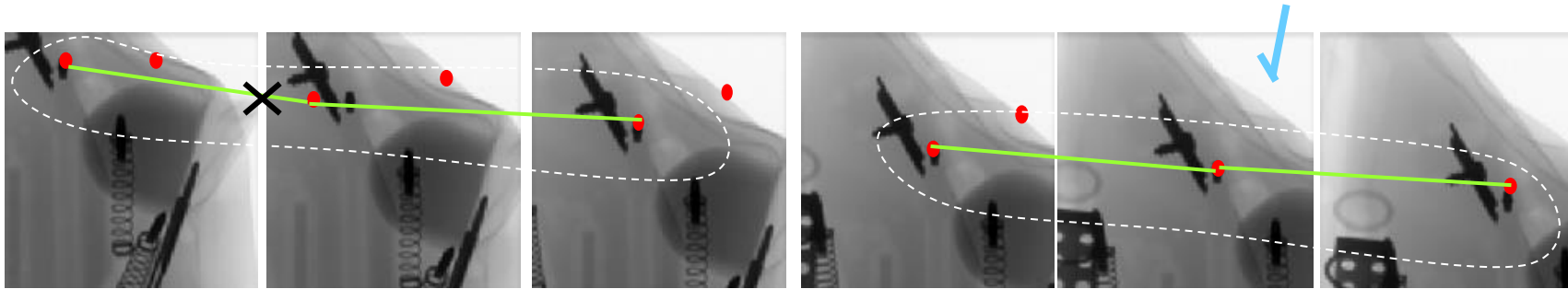
Tracking algorithm



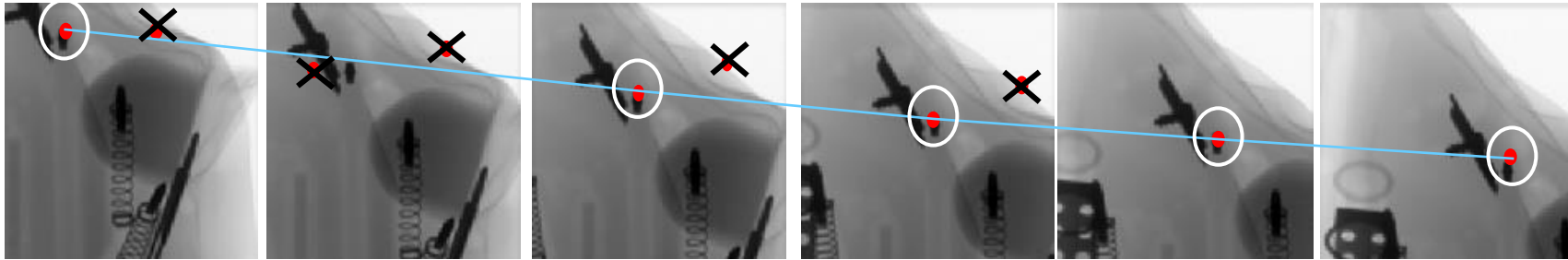
Tracking algorithm



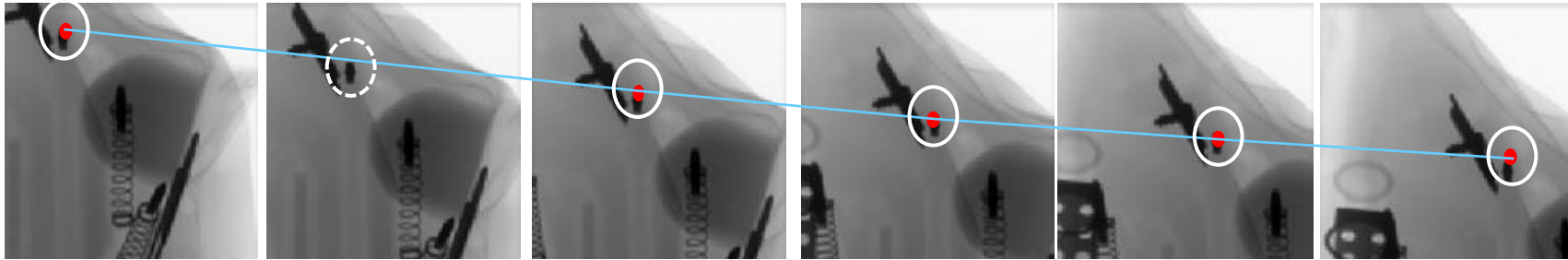
Tracking algorithm



Tracking algorithm

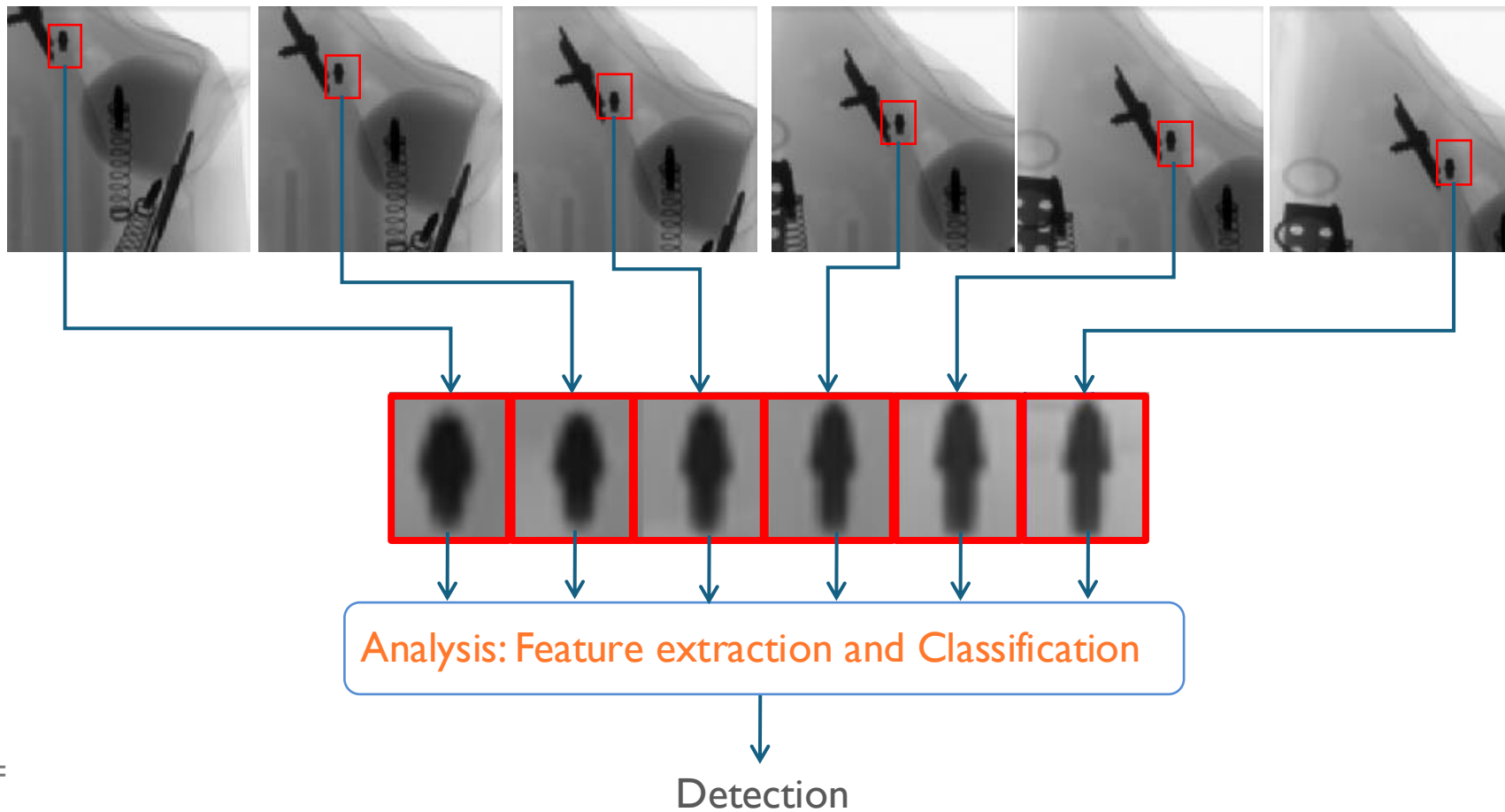


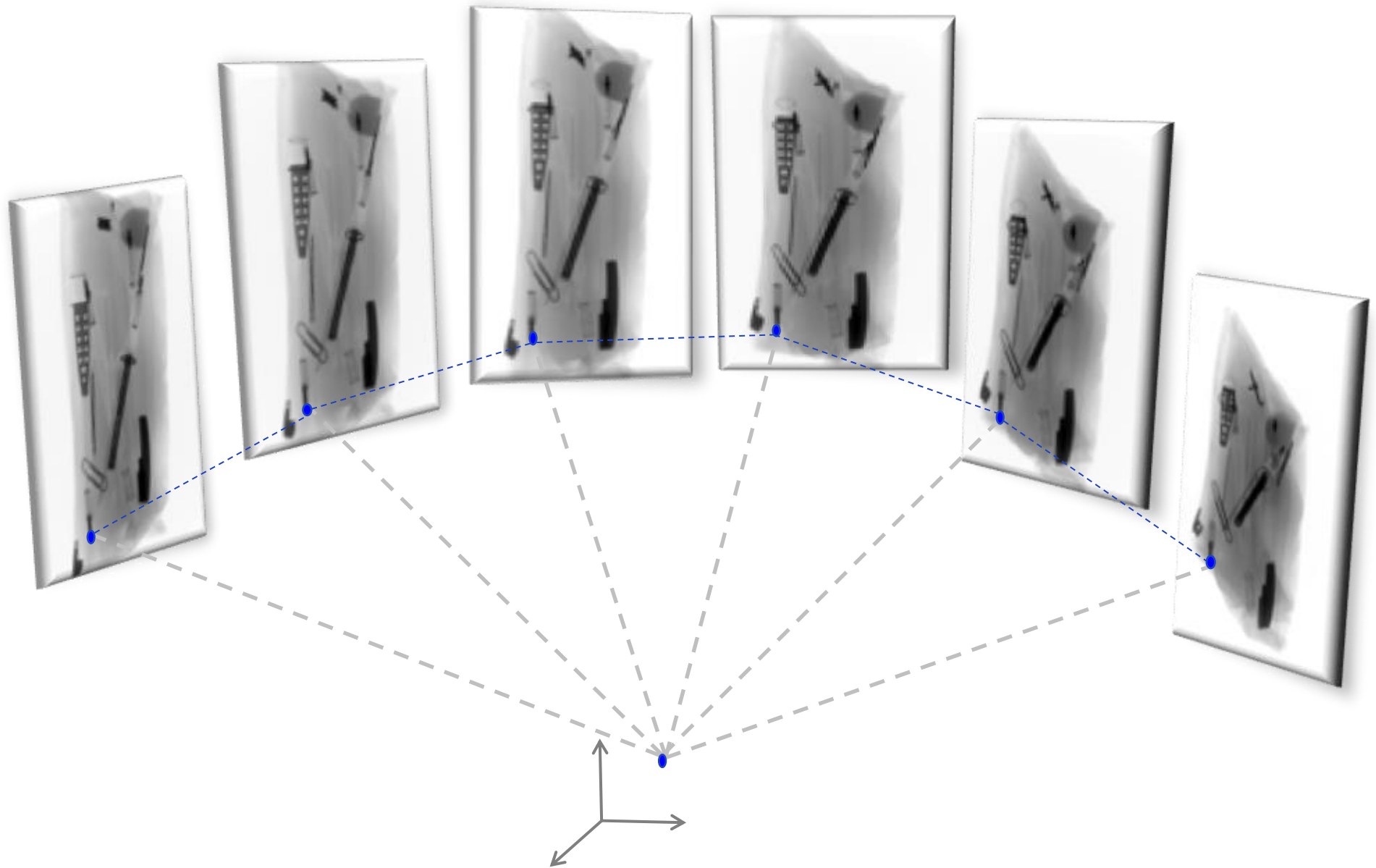
Tracking algorithm

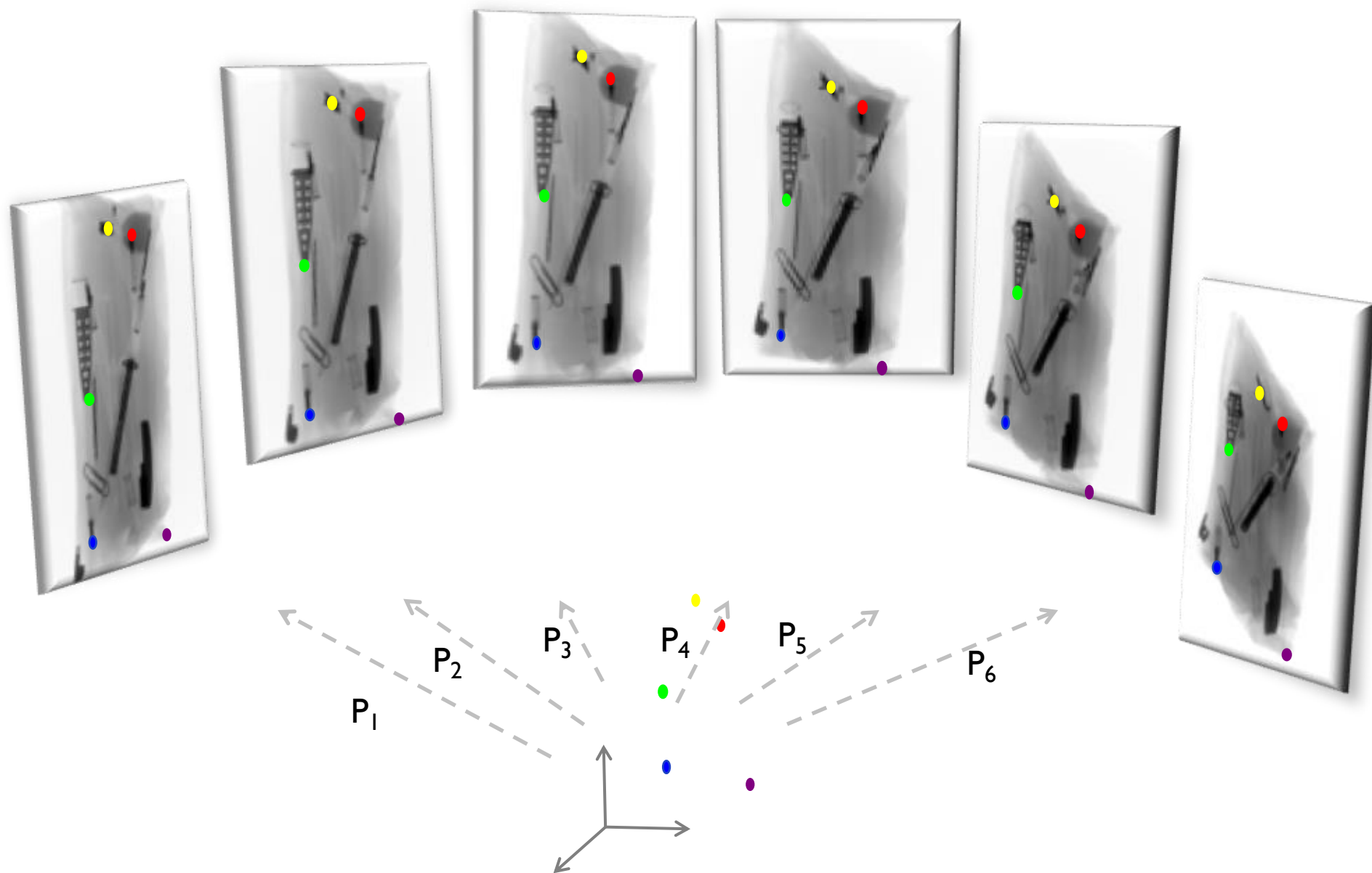


- It is not required that the object parts have to be segmented in all views because the missing views can be reconstructed.
- It is allowed to have false segmentations because they are filtered out.

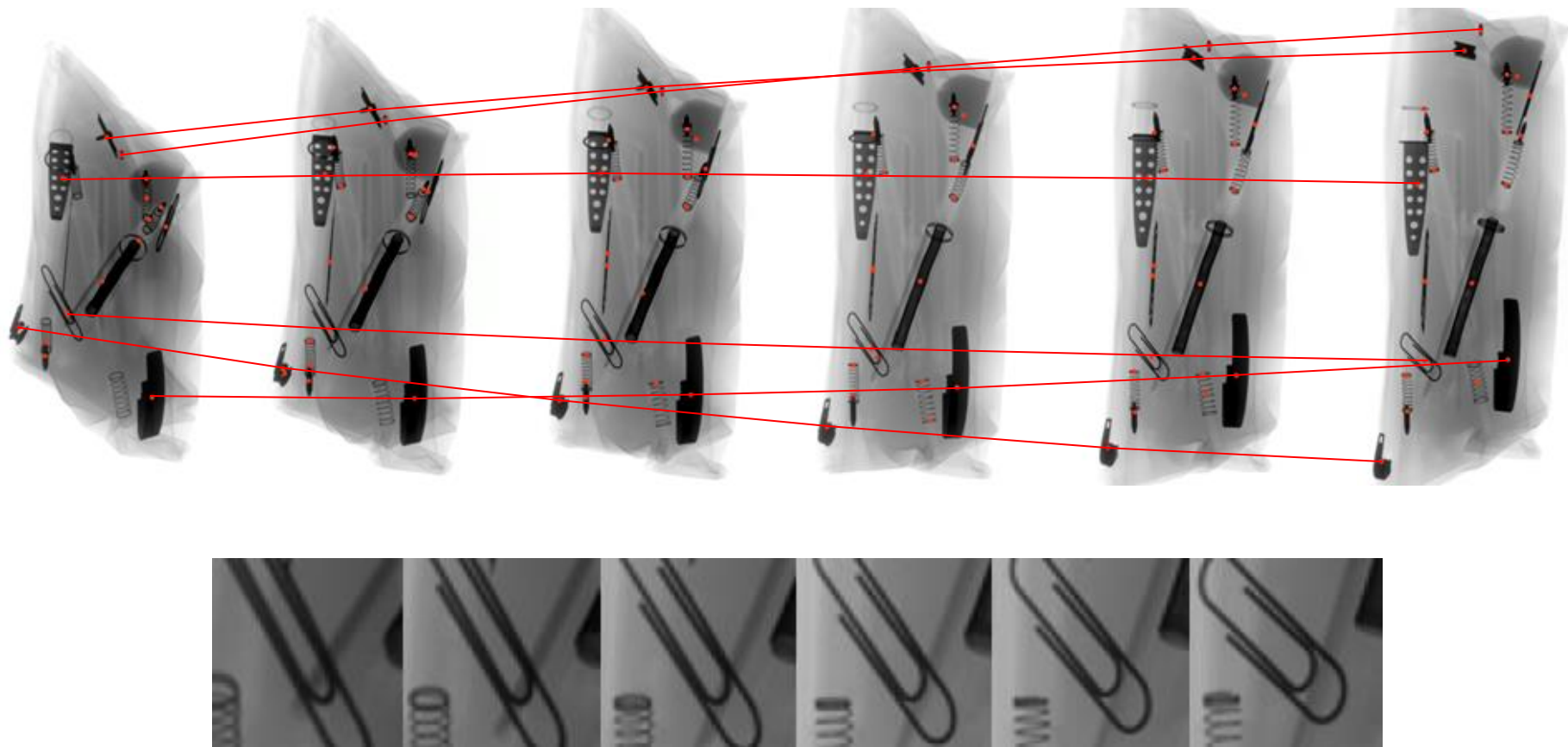
Tracking algorithm



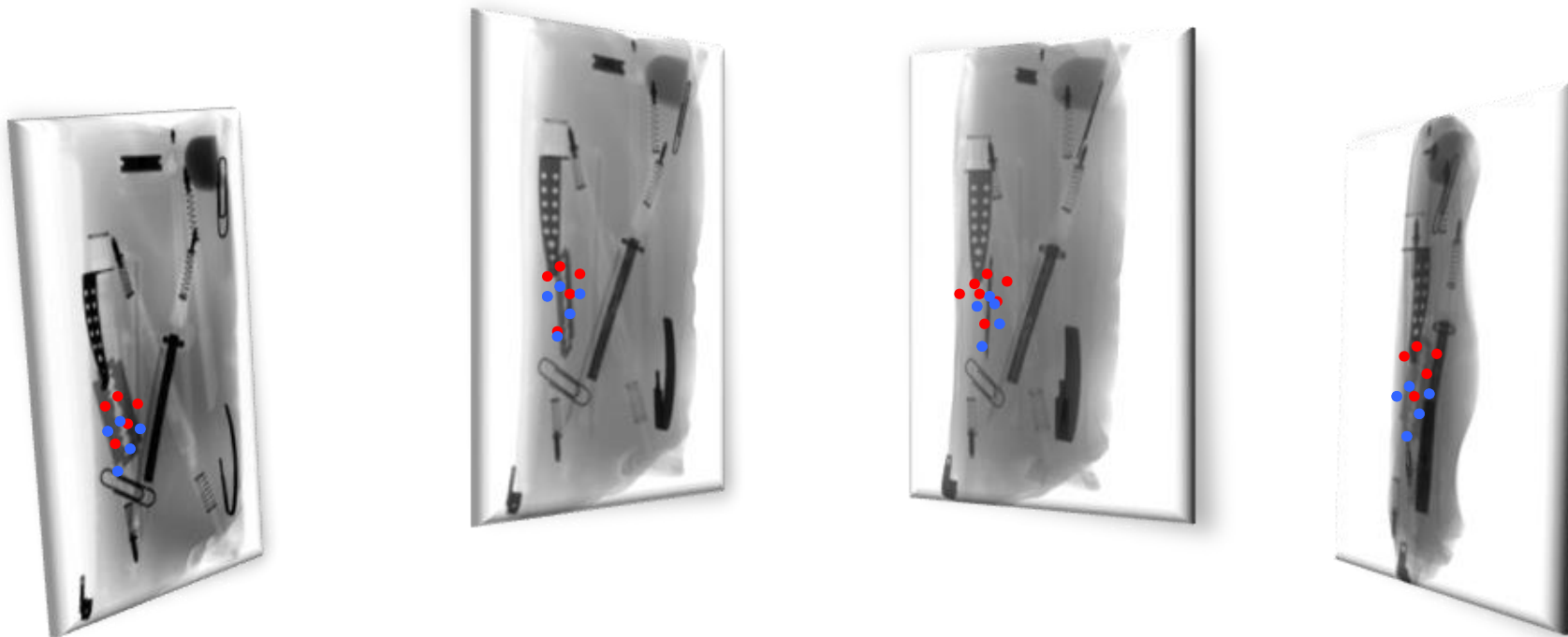


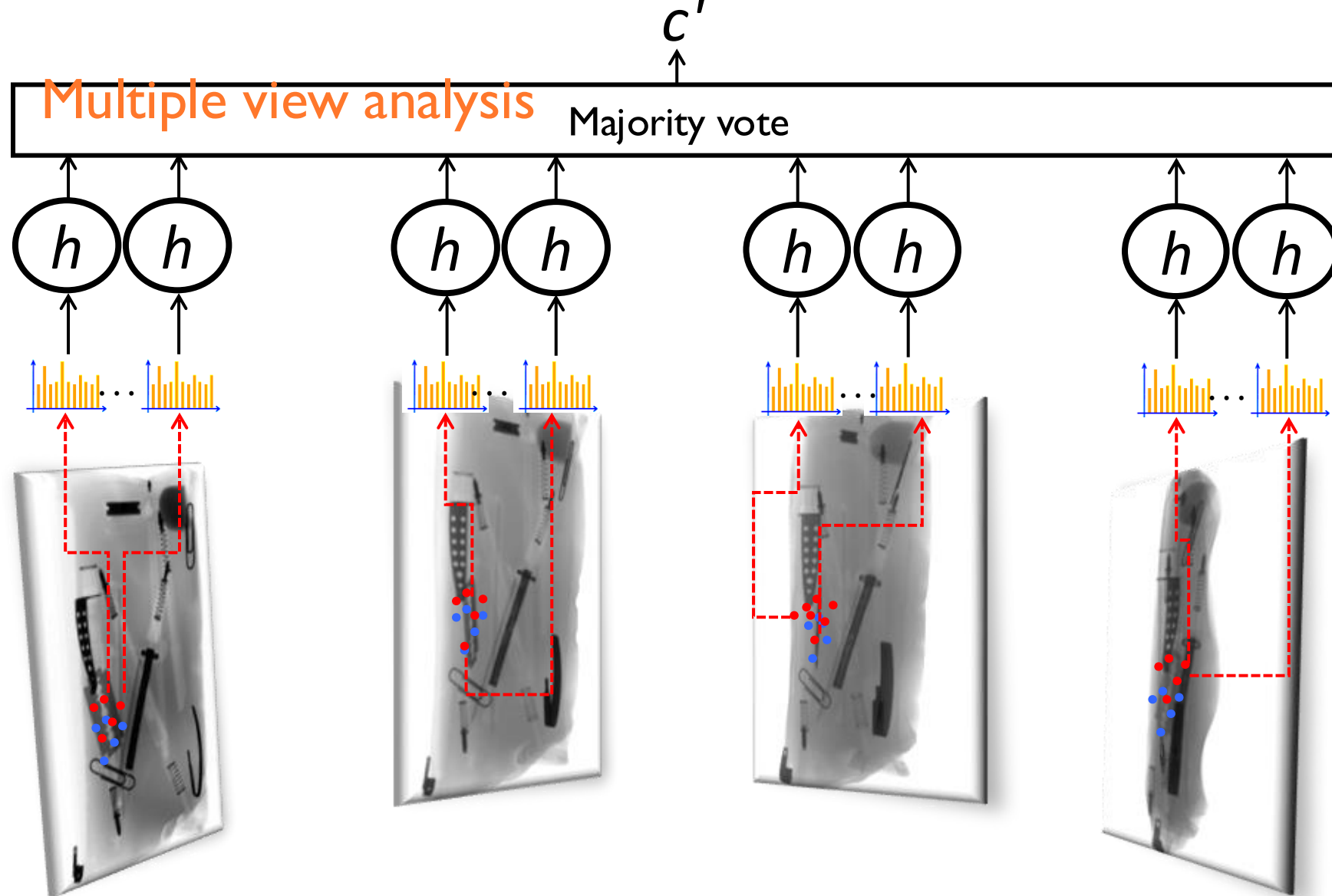


X-ray Testing: Multiple view analysis

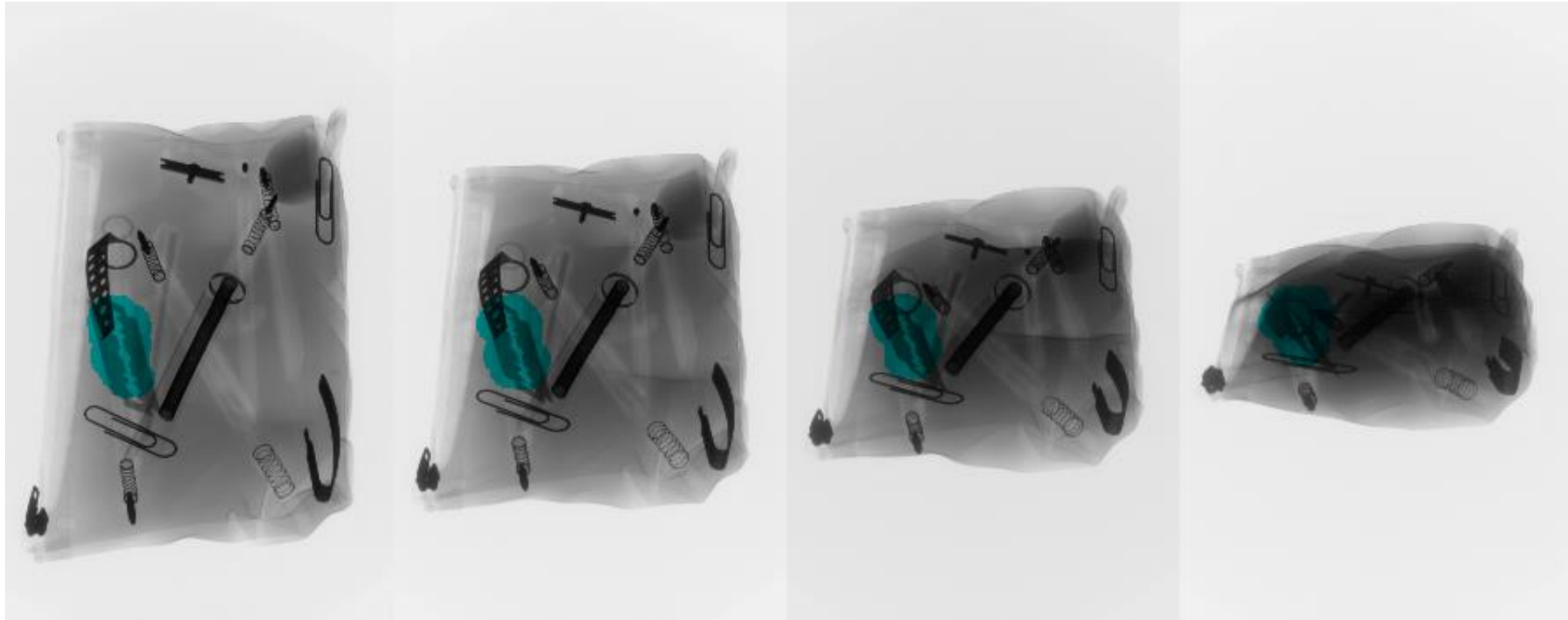


Multiple view analysis



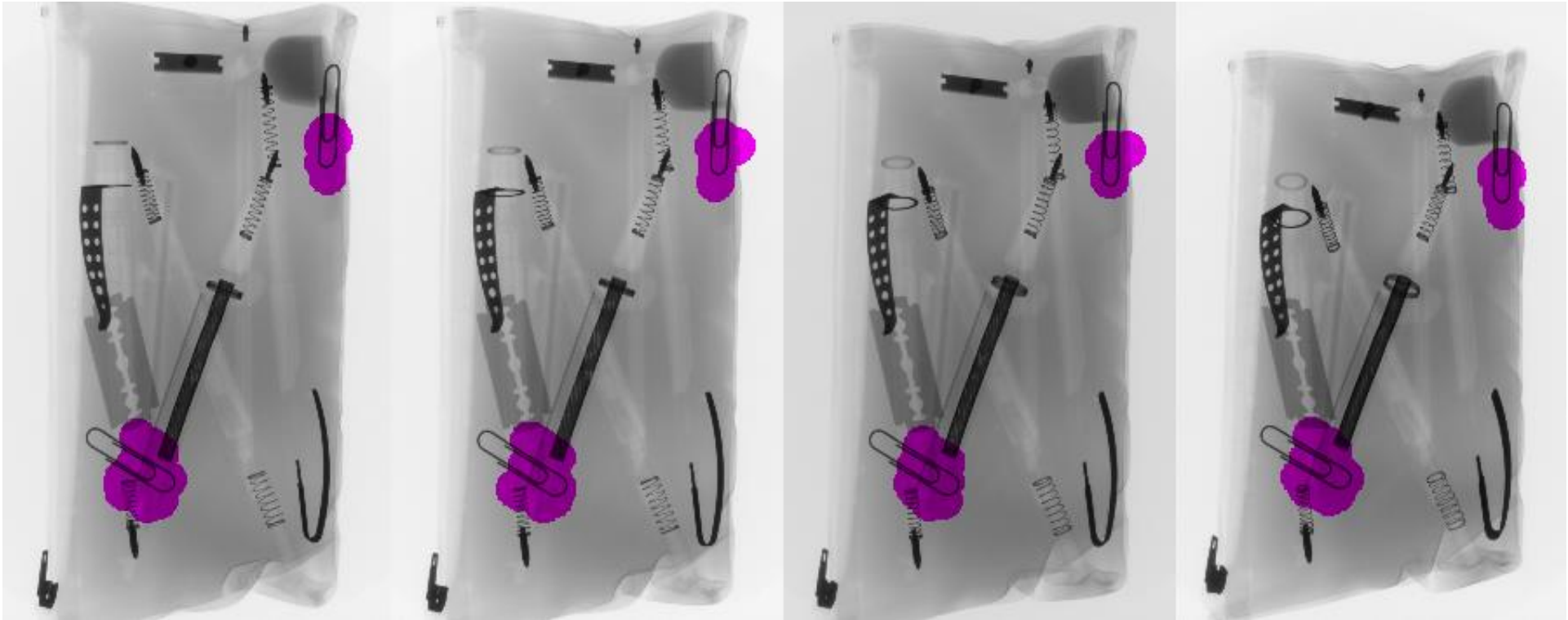


Multiple view analysis: Experimental Results



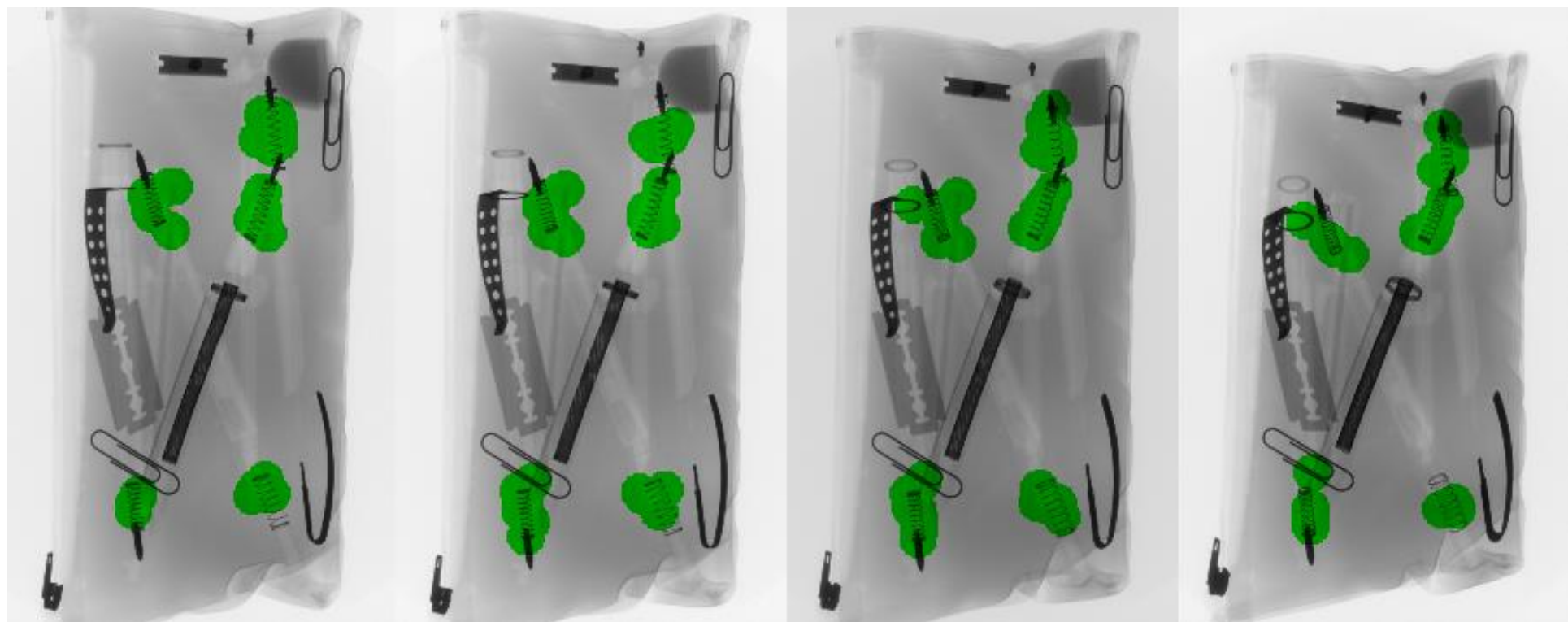
[BLADE]

Multiple view analysis: Experimental Results



[CLIPS]

Multiple view analysis: Experimental Results



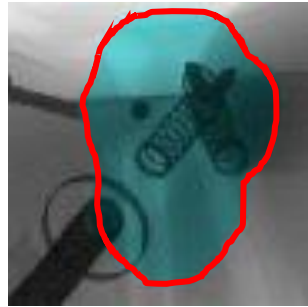
[SPRINGS]

Multiple view analysis: Experimental Results

Recognition with some degree of occlusion



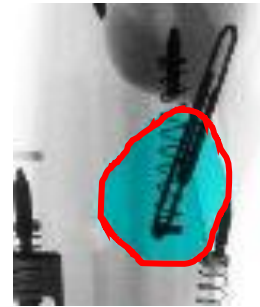
[SPRING]



[SPRINGS]



[CLIP]

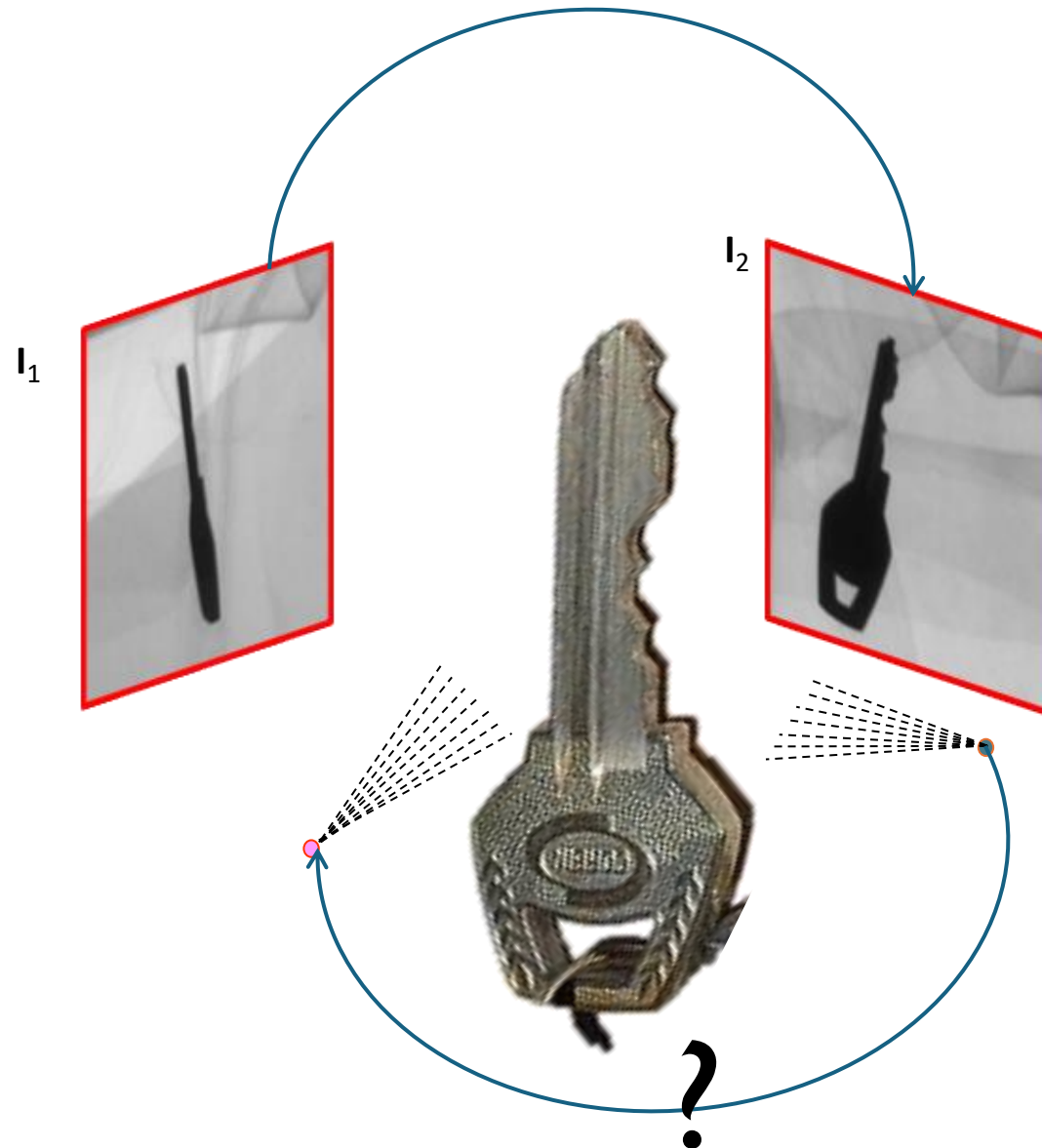


[CLIP]



[BLADE]

Active Vision: Principle



Active Vision: Algorithm

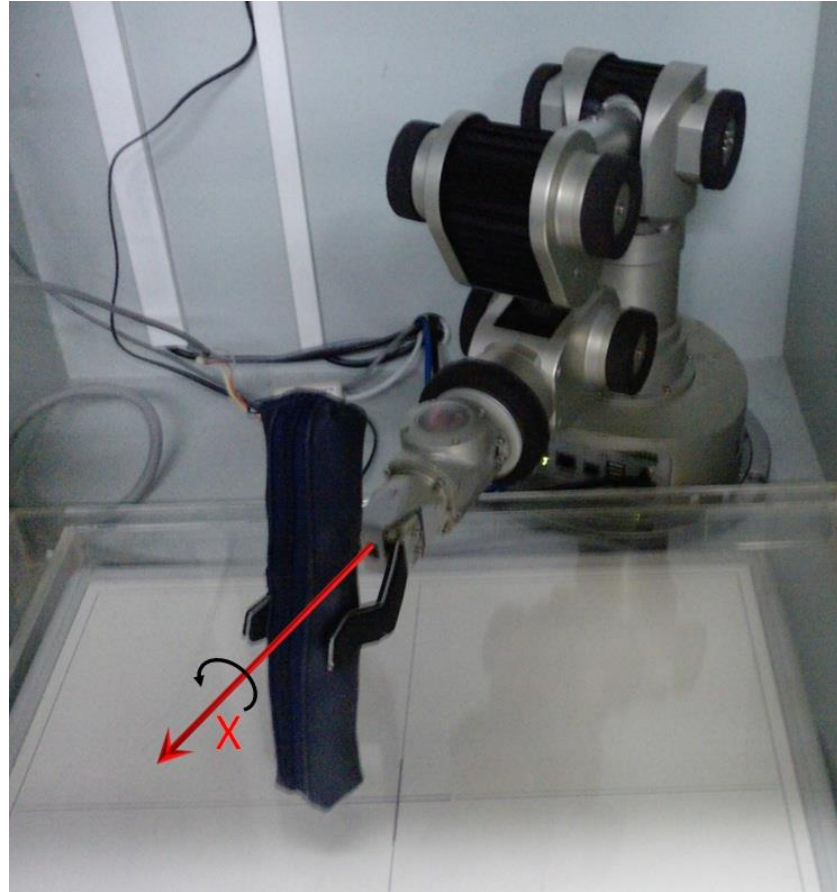
1. Number of Images = 0
Position = Random Position
2. Repeat until Number of Images $< N$
3. Move inspection object to Position
4. Acquisition of a new X-ray image
Number of Images = Number Images + 1
5. Single view detection:

[DETECTION] End

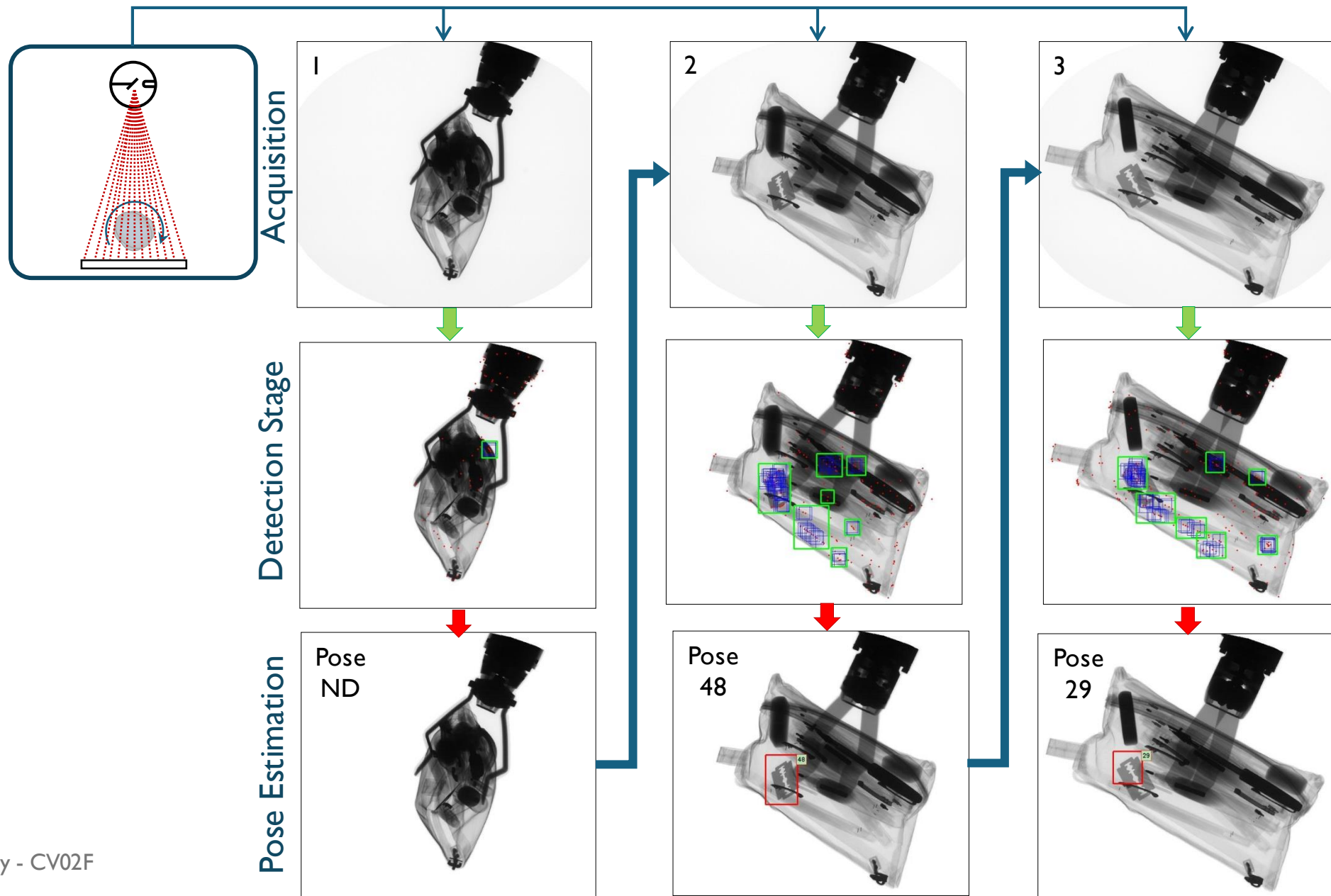
[NOTHING] Position = Position + Θ
Go to step 2

[COULD BE] Pose estimation
Estimation of Better Position
Position = Better Position
Go to step 2

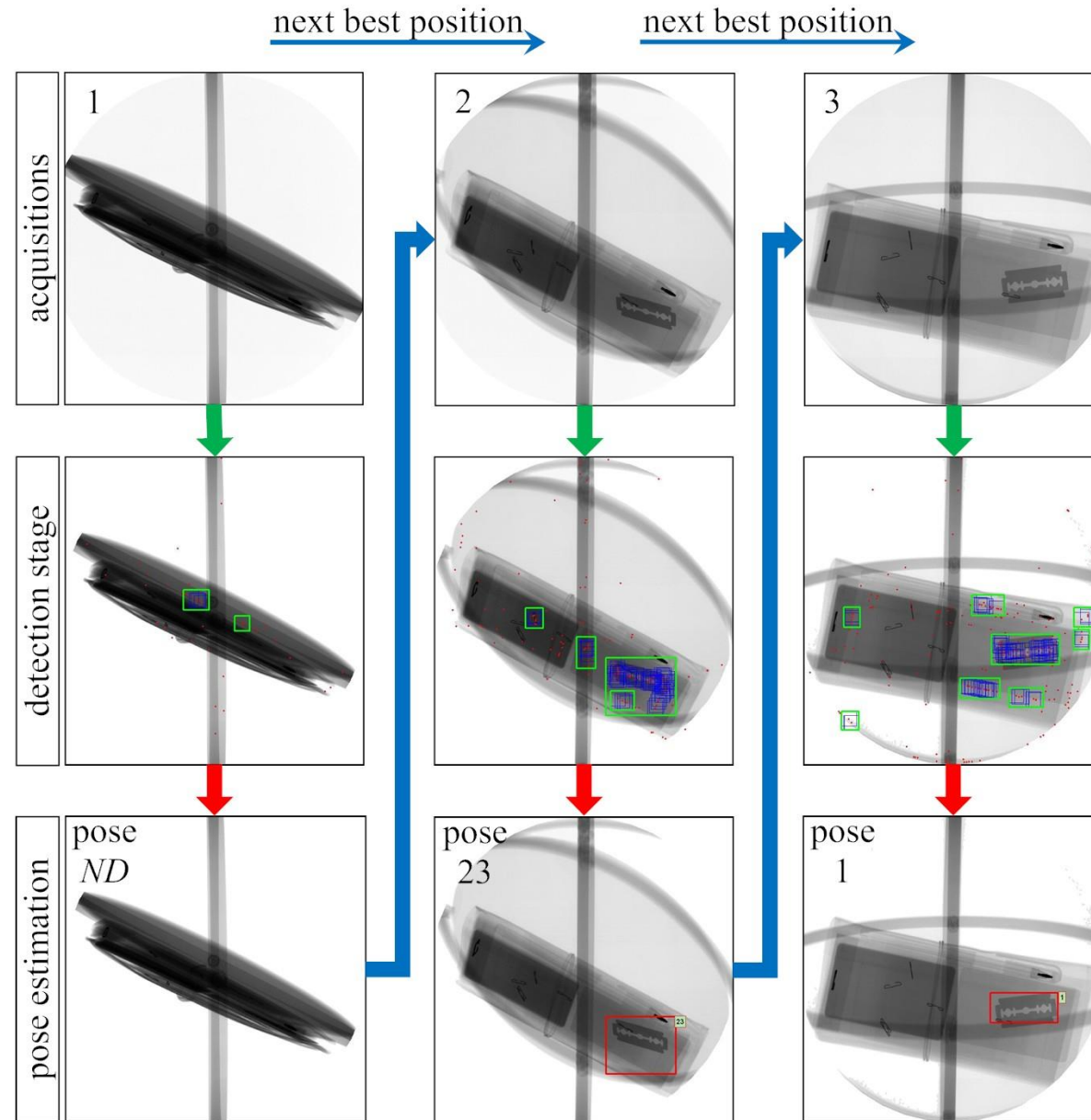
Active Vision: Robotic system



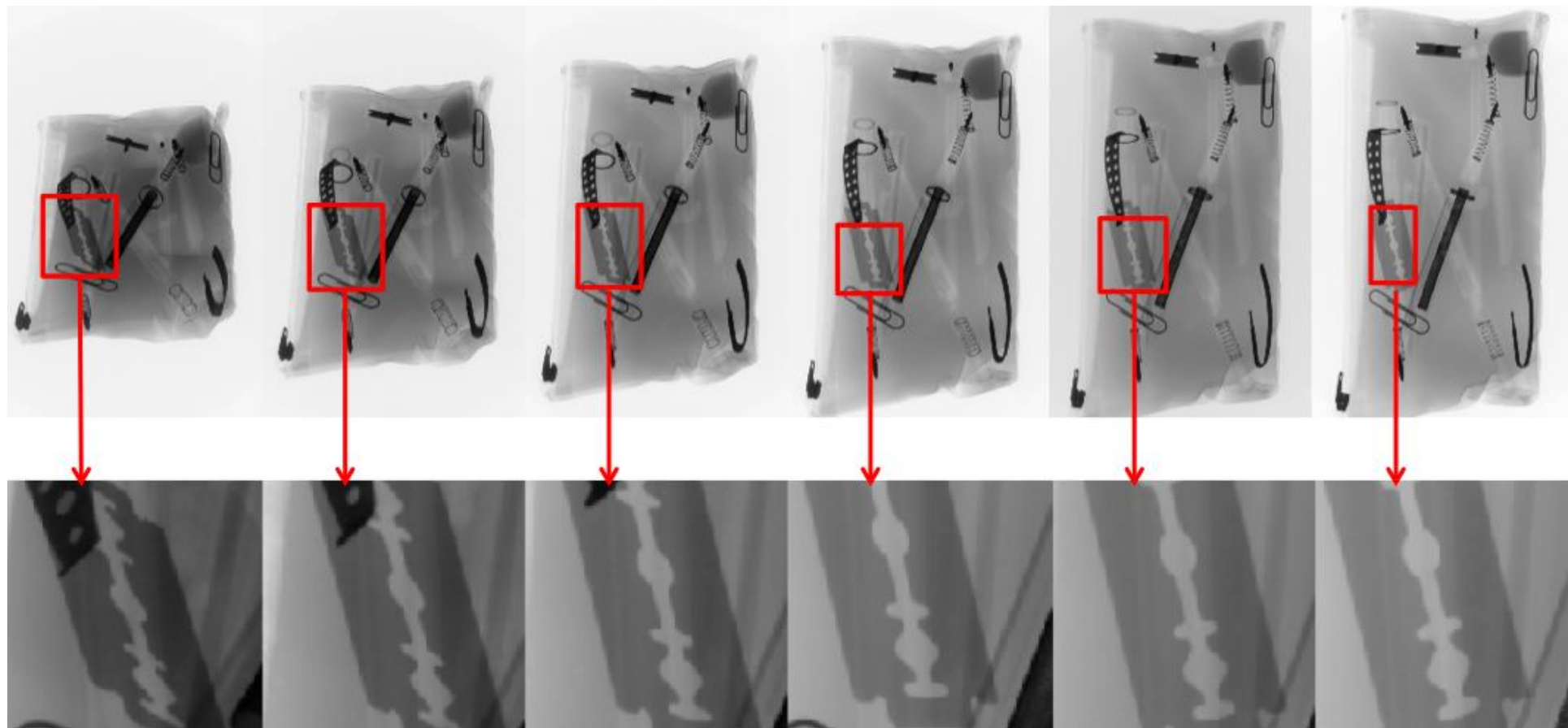
Active Vision: Experimental results



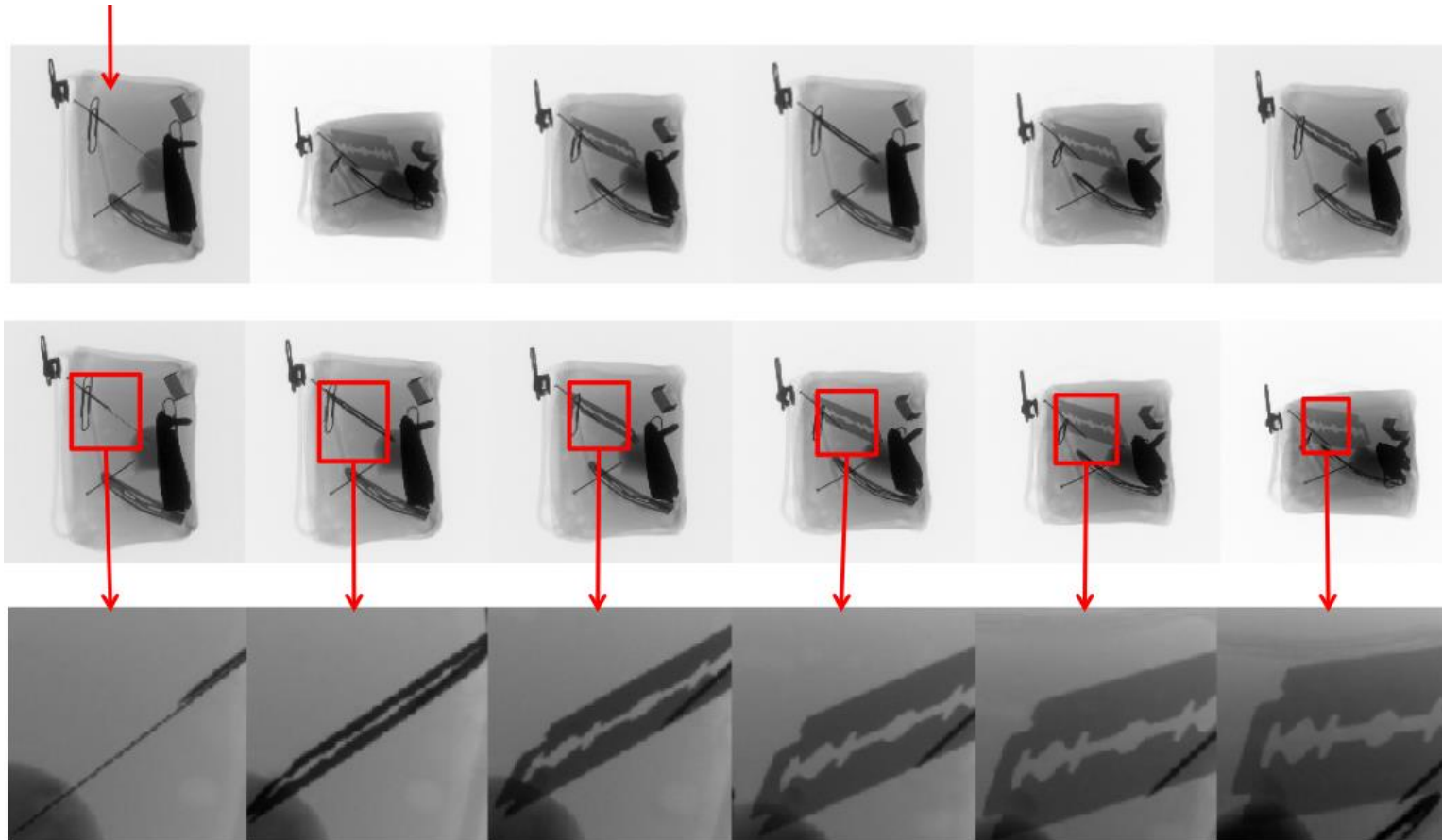
Active Vision: Experimental results



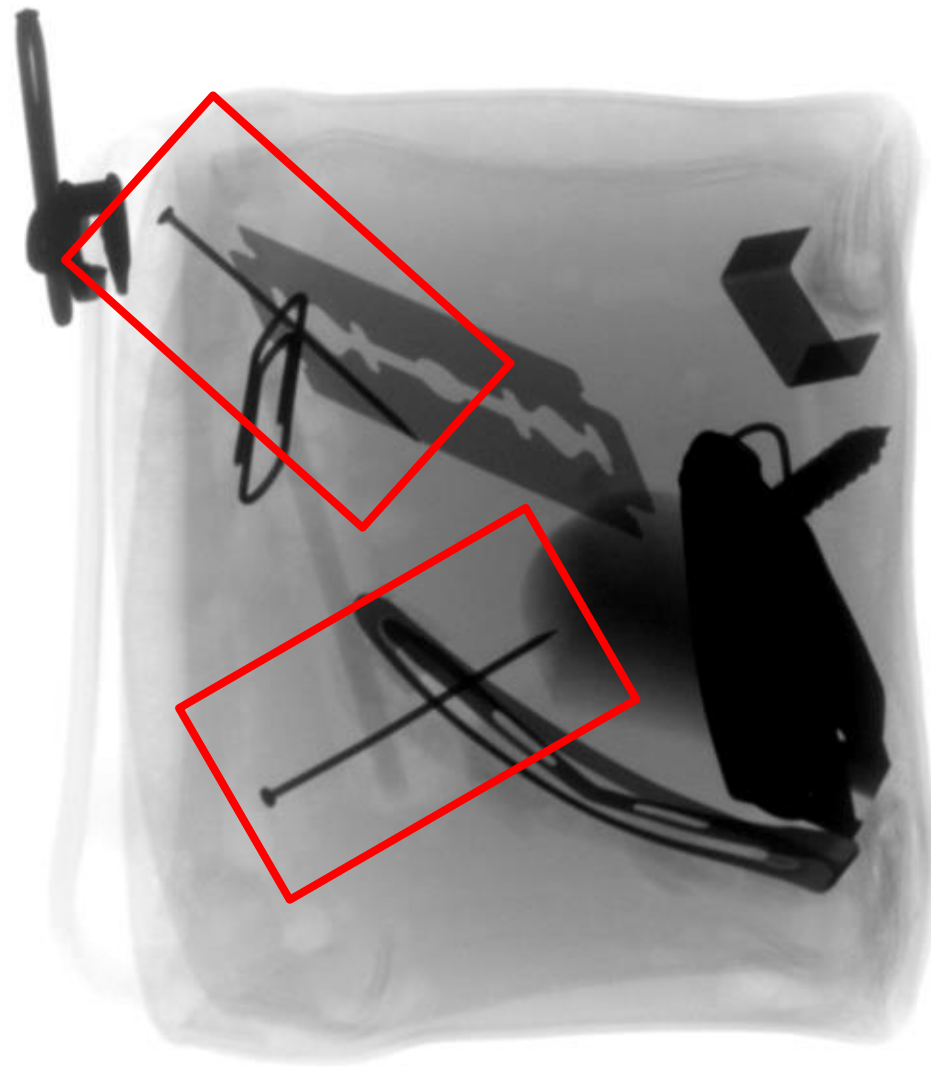
More Examples



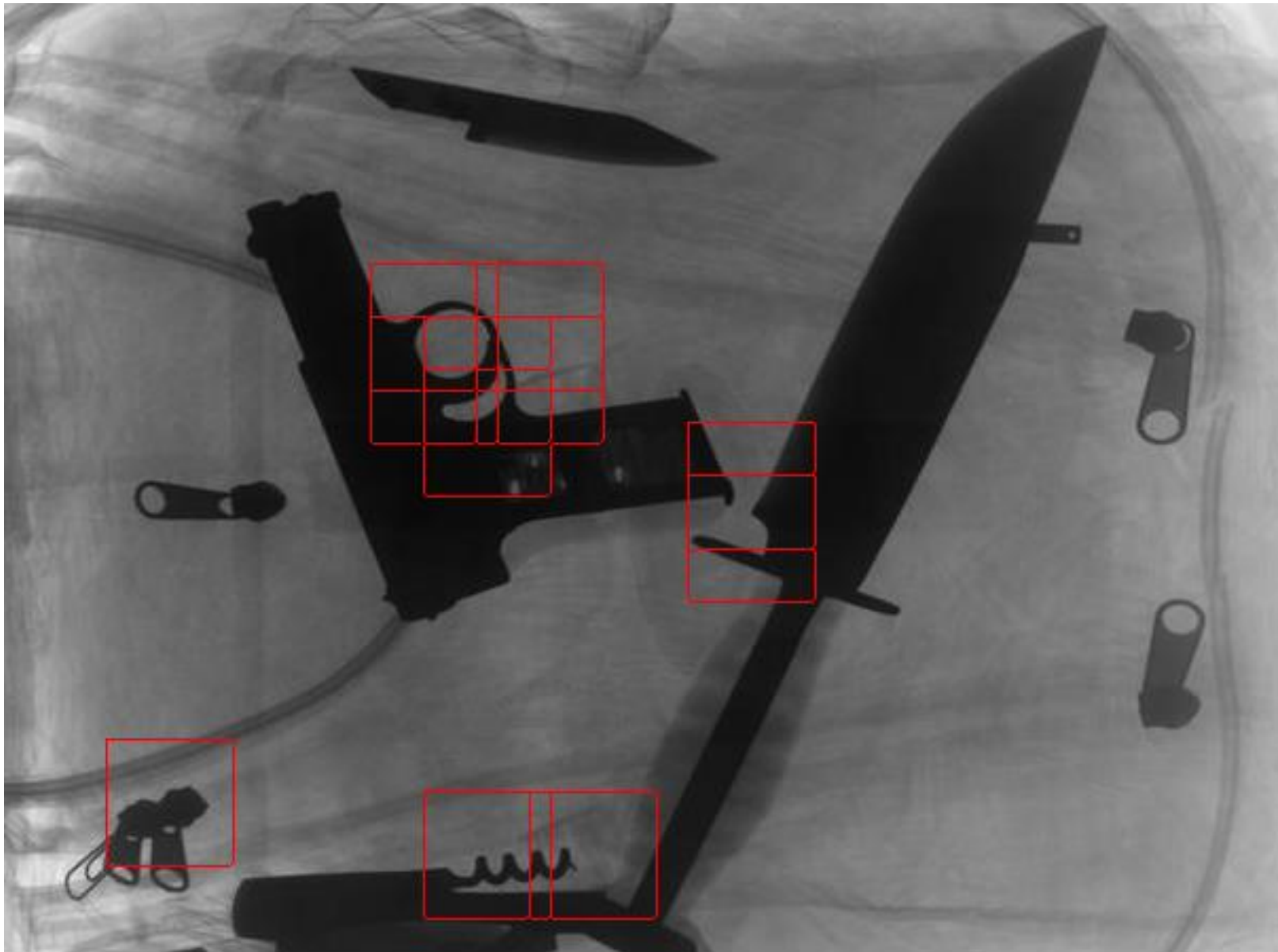
Detection of a razor blade



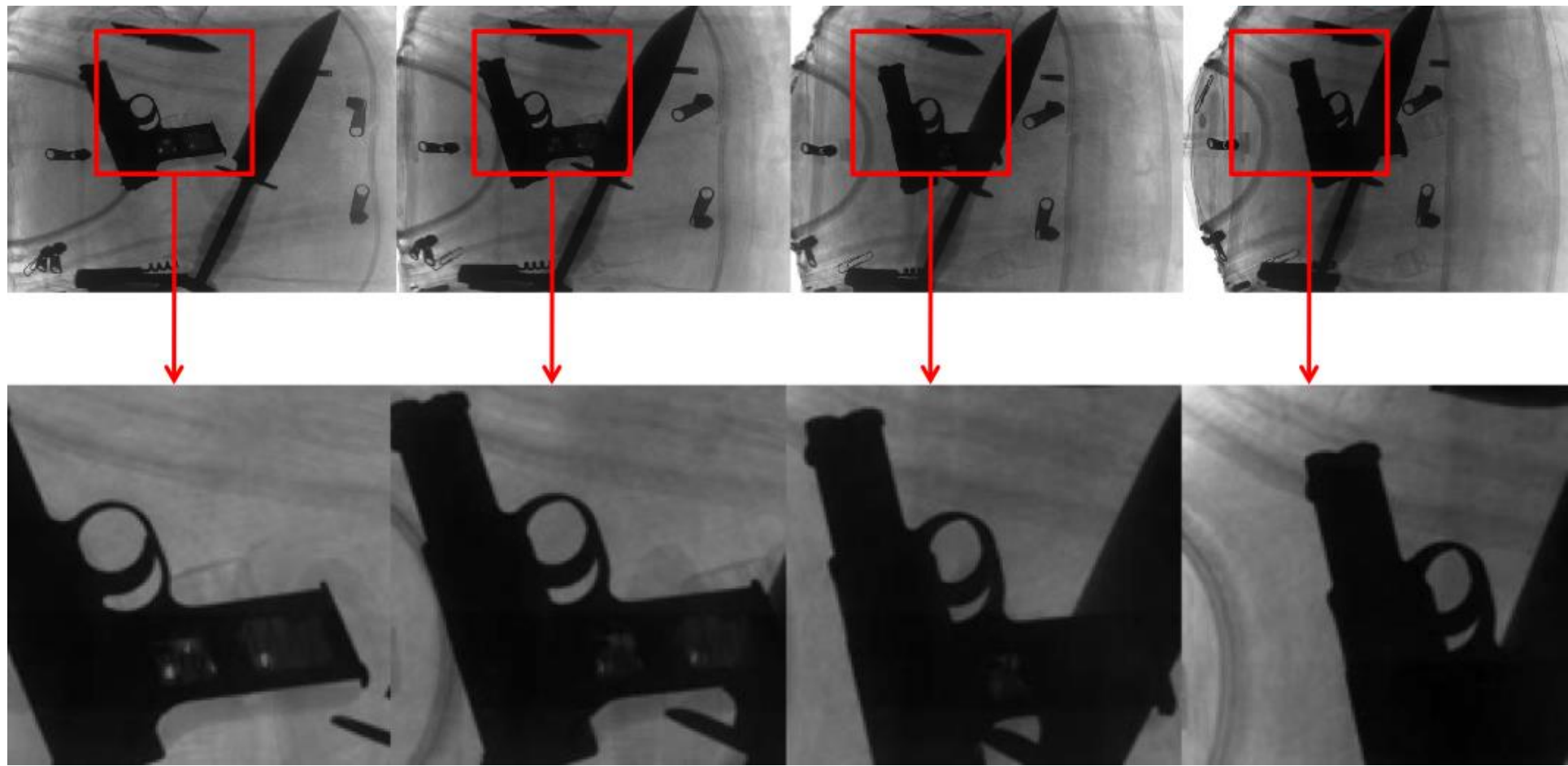
Detection of a razor blade



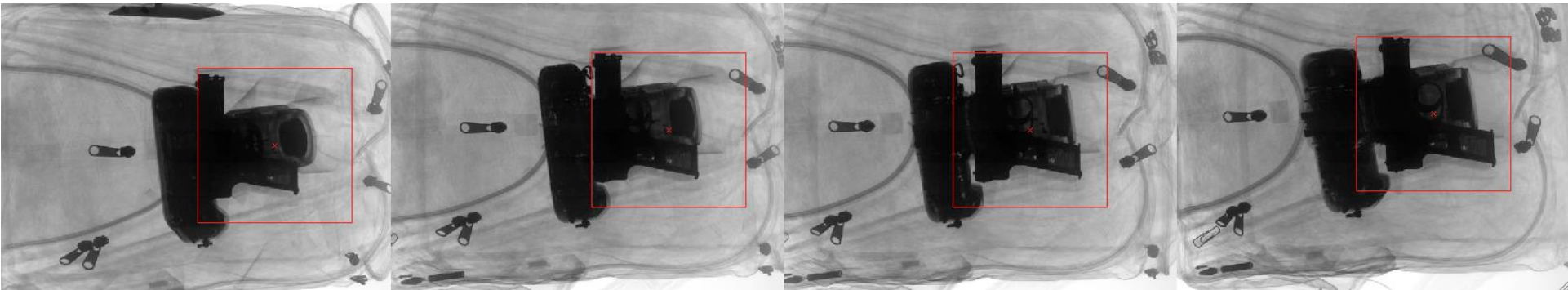
Detection of pins



Detection of guns (mono detection)

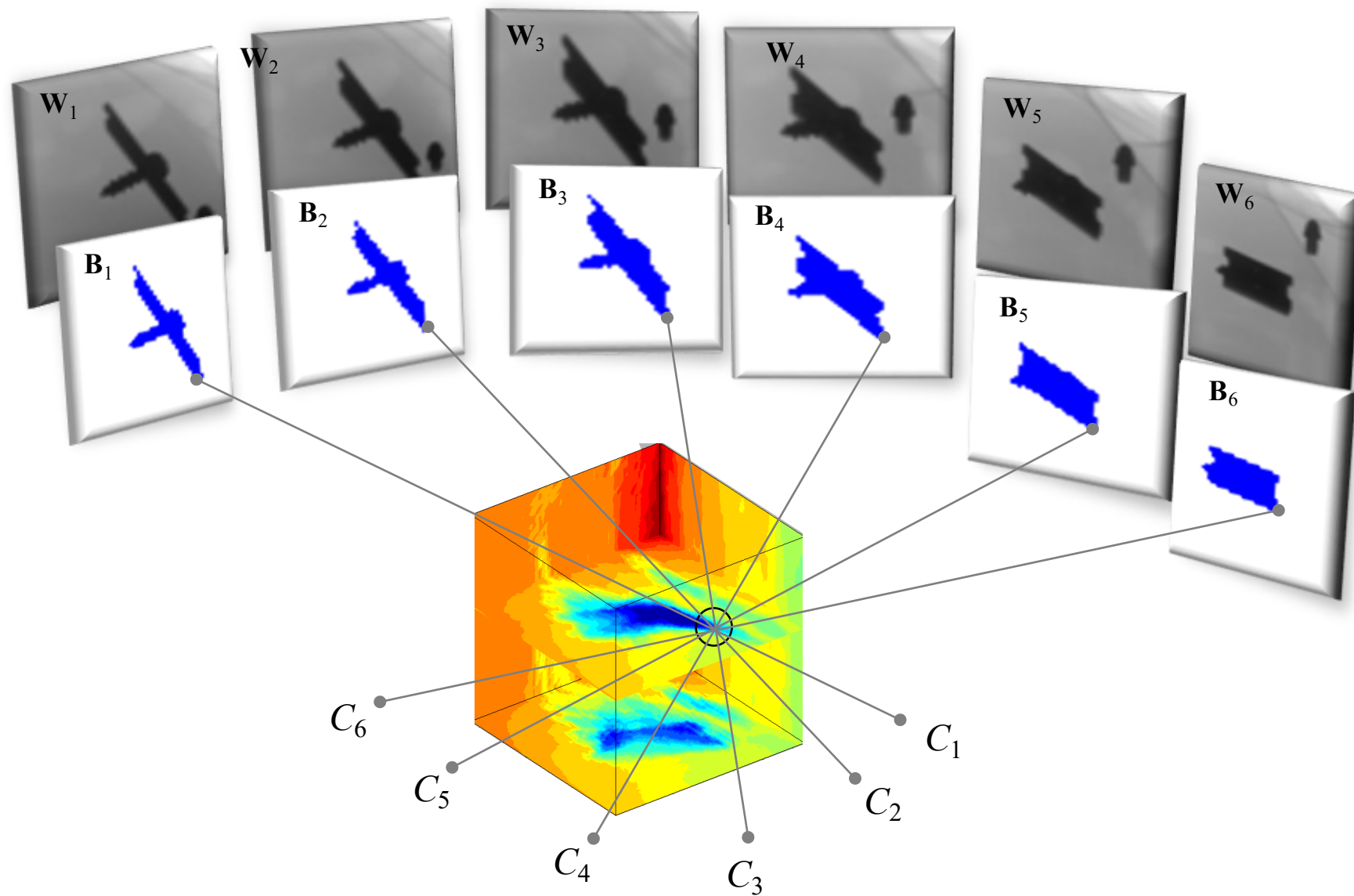


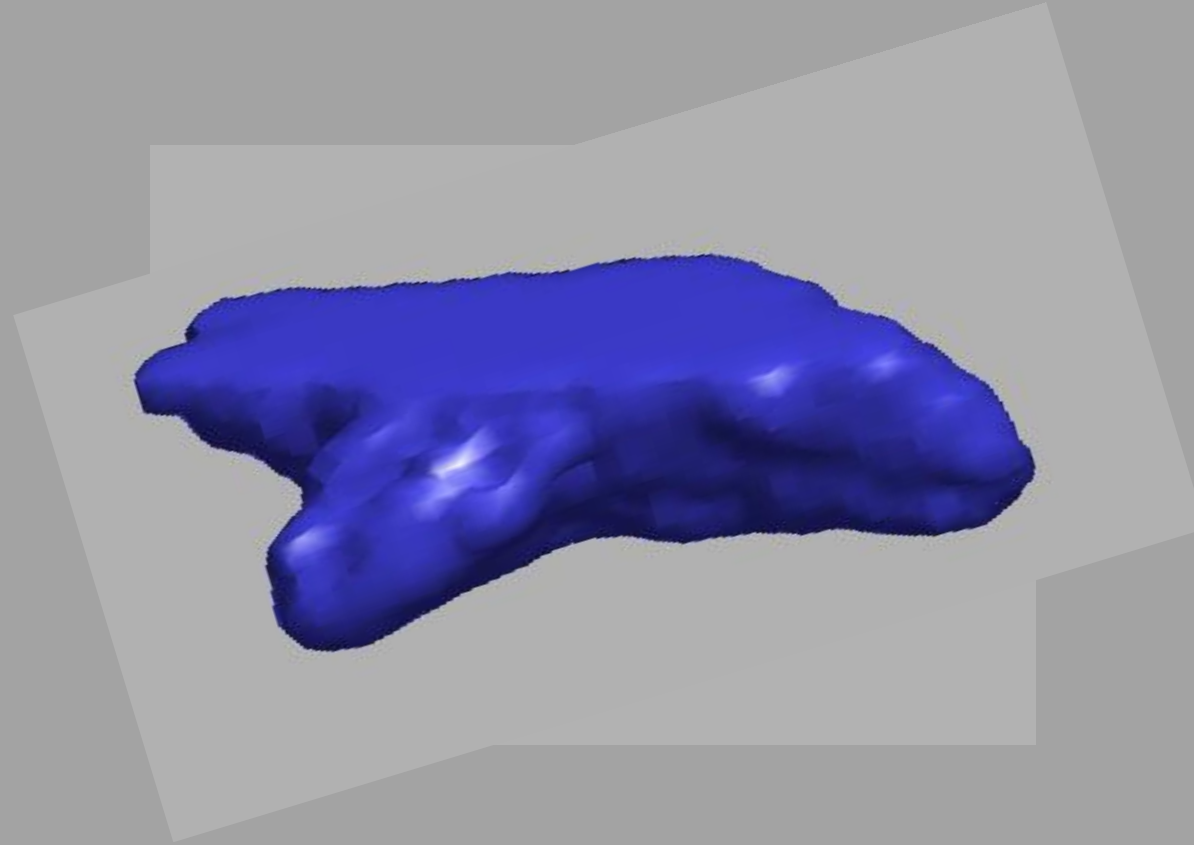
Detection of guns (multiple view)

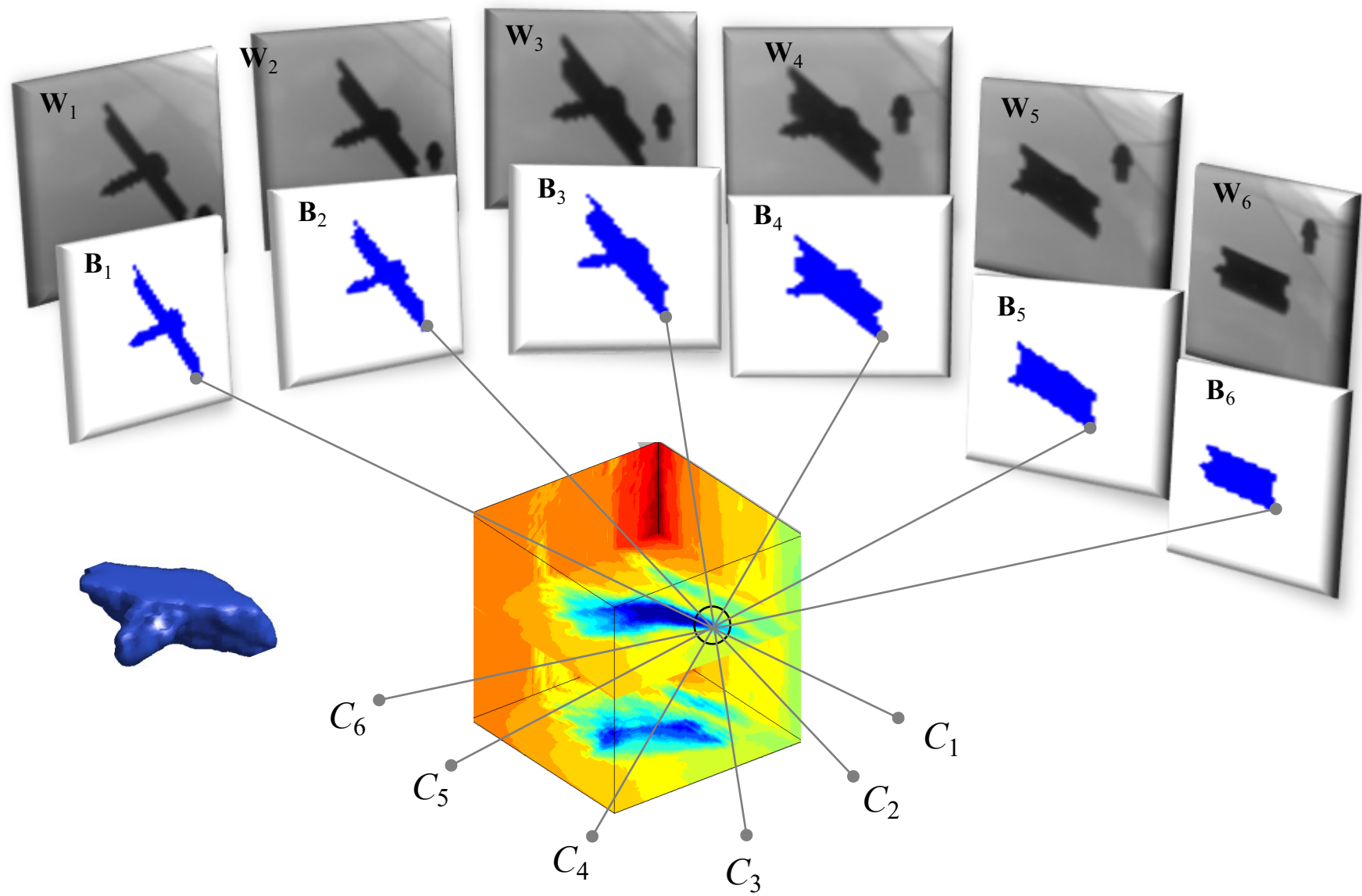


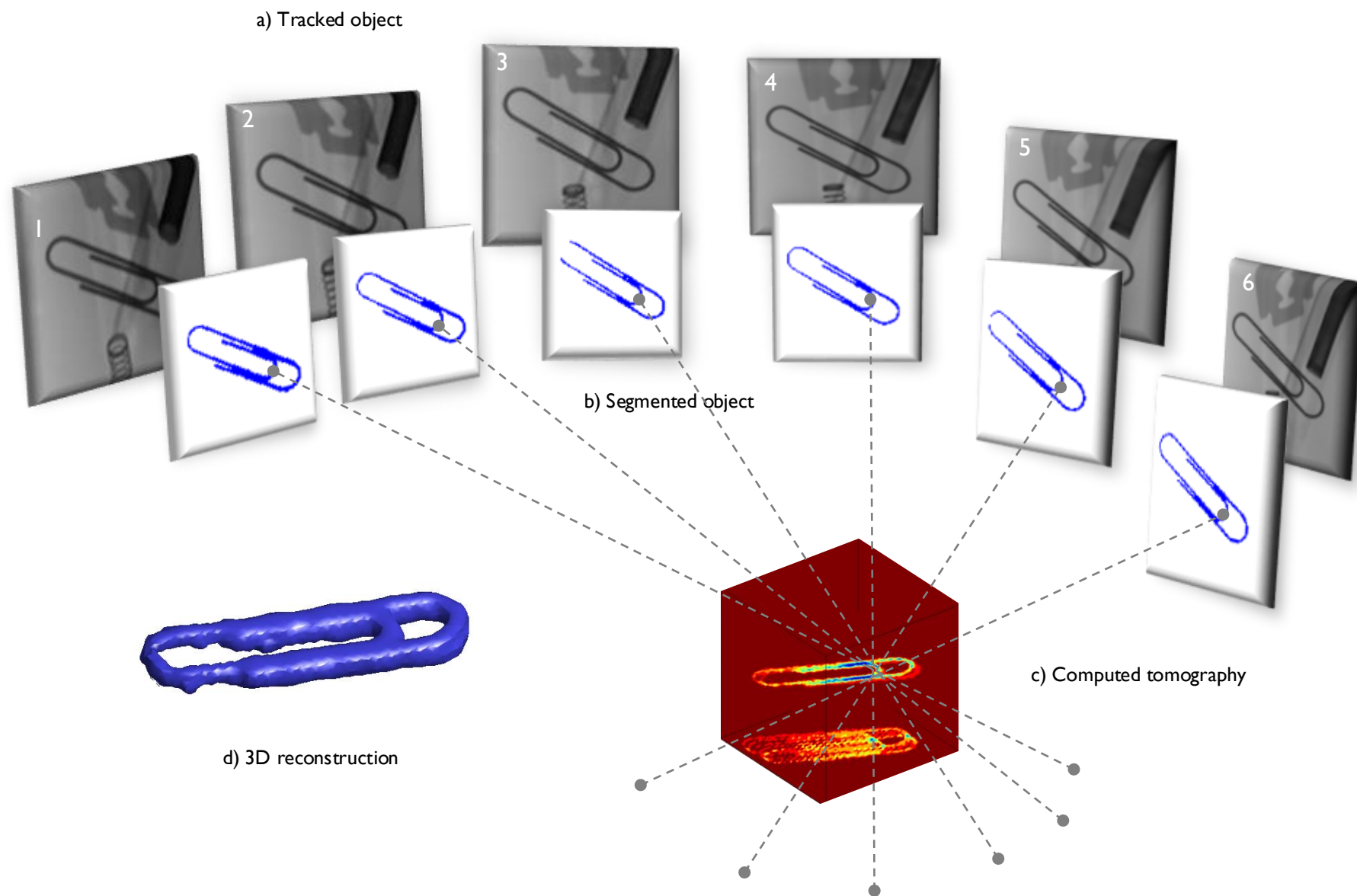
What can we do with a sequence?











Images

Load seq_guns2.mat

Show 1/4 - +

Object

Search ok

Select 1/2 - +

☒ All objects

☐ One object

☒ All frames

Tracking

None ▾

☐ Structure

☐ Details

☐ Multi

Tomography

Go

Backprojection ▾

X Y Z

slice [64 0 0]

3D [12 4 64]

